

Numerical Methods

CHAPTER CONTENTS

9.1 LU-Decompositions 509

9.2 The Power Method 519

9.3 Comparison of Procedures for Solving Linear Systems 528

9.4 Singular Value Decomposition 532

9.5 Data Compression Using Singular Value Decomposition 540

Introduction

This chapter is concerned with “numerical methods” of linear algebra, an area of study that encompasses techniques for solving large-scale linear systems and for finding numerical approximations of various kinds. It is not our objective to discuss algorithms and technical issues in fine detail since there are many excellent books on the subject. Rather, we will be concerned with introducing some of the basic ideas and exploring two important contemporary applications that rely heavily on numerical ideas—singular value decomposition and data compression. A computing utility such as MATLAB, *Mathematica*, or Maple is recommended for Sections 9.2 to 9.5.

9.1 LU-Decompositions

Up to now, we have focused on two methods for solving linear systems, Gaussian elimination (reduction to row echelon form) and Gauss–Jordan elimination (reduction to reduced row echelon form). While these methods are fine for the small-scale problems in this text, they are not suitable for large-scale problems in which computer roundoff error, memory usage, and speed are concerns. In this section we will discuss a method for solving a linear system of n equations in n unknowns that is based on factoring its coefficient matrix into a product of lower and upper triangular matrices. This method, called “LU-decomposition,” is the basis for many computer algorithms in common use.

Solving Linear Systems by Factoring

Our first goal in this section is to show how to solve a linear system $A\mathbf{x} = \mathbf{b}$ of n equations in n unknowns by factoring the coefficient matrix A . We begin with some terminology.

Definition 1

A factorization of a square matrix A as

$$A = LU \quad (1)$$

where the matrix L is lower triangular and the matrix U is upper triangular, is called an ***LU-decomposition*** (or ***LU-factorization***) of A .

Before we consider the problem of obtaining an LU -decomposition, we will explain how such decompositions can be used to solve linear systems, and we will give an illustrative example.

The Method of LU -Decomposition

Step 1. Rewrite the system $A\mathbf{x} = \mathbf{b}$ as

$$LU\mathbf{x} = \mathbf{b} \quad (2)$$

Step 2. Make the substitution

$$\mathbf{y} = U\mathbf{x} \quad (3)$$

then rewrite (2) as $L\mathbf{y} = \mathbf{b}$ and solve this system for $\mathbf{y}$.

Step 3. Substitute $\mathbf{y}$ in (3) and solve for $\mathbf{x}$.

This procedure, which is illustrated in **Figure 9.1.1**, replaces the single linear system $A\mathbf{x} = \mathbf{b}$ by a pair of linear systems

$$\begin{aligned} U\mathbf{x} &= \mathbf{y} \\ L\mathbf{y} &= \mathbf{b} \end{aligned}$$

that must be solved in succession. However, since each of these systems has a triangular coefficient matrix, it generally turns out to involve no more computation to solve the two systems than to solve the original system directly.

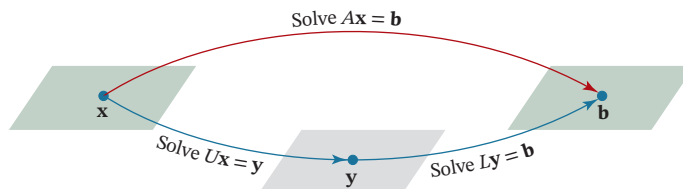


FIGURE 9.1.1

Historical Note

In 1979 an important library of machine-independent linear algebra programs called LINPACK was developed at Argonne National Laboratories. Many of the programs in that library use the decomposition methods that we will study in this section. Variations of the LINPACK routines are used in many computer programs, including MATLAB, *Mathematica*, and Maple.

EXAMPLE 1 | Solving $Ax = b$ by LU-Decomposition

Later in this section we will derive the factorization

$$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \quad (4)$$

$A \qquad = \qquad L \qquad U$

Use this result to solve the linear system

$$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

$A \qquad \mathbf{x} = \mathbf{b}$

From (4) we can rewrite this system as

$$\begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \quad (5)$$

$L \qquad U \qquad \mathbf{x} = \mathbf{b}$

As specified in Step 2 above, let us define y_1, y_2 , and y_3 by the equation

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \quad (6)$$

$U \qquad \mathbf{x} = \mathbf{y}$

which allows us to rewrite (5) as

$$\begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \quad (7)$$

$L \qquad \mathbf{y} = \mathbf{b}$

or equivalently as

$$\begin{aligned} 2y_1 &= 2 \\ -3y_1 + y_2 &= 2 \\ 4y_1 - 3y_2 + 7y_3 &= 3 \end{aligned}$$

This system can be solved by a procedure that is similar to back substitution, except that we solve the equations from the top down instead of from the bottom up. This procedure, called **forward substitution**, yields

$$y_1 = 1, \quad y_2 = 5, \quad y_3 = 2$$

(verify). As indicated in Step 3 above, we substitute these values into (6), which yields the linear system

$$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$$

or, equivalently,

$$\begin{aligned}x_1 + 3x_2 + x_3 &= 1 \\x_2 + 3x_3 &= 5 \\x_3 &= 2\end{aligned}$$

Solving this system by back substitution yields

$$x_1 = 2, \quad x_2 = -1, \quad x_3 = 2$$

(verify).

Finding LU -Decompositions

The preceding example illustrates that once an LU -decomposition of A is obtained, a linear system $A\mathbf{x} = \mathbf{b}$ can be solved by one forward substitution and one backward substitution. The main advantage of this method over Gaussian and Gauss–Jordan elimination is that it “decouples” A from $\mathbf{b}$ so that for solving a *sequence* of linear systems with the same coefficient matrix A , say

$$A\mathbf{x} = \mathbf{b}_1, \quad A\mathbf{x} = \mathbf{b}_2, \dots, \quad A\mathbf{x} = \mathbf{b}_k$$

the work in factoring A need only be performed once, after which it can be reused for each system in the sequence. Such sequences occur in problems in which the matrix A remains fixed but the vector $\mathbf{b}$ varies with time.

Historical Note



**Alan Mathison
Turing**
(1912–1954)

Although the ideas were known earlier, credit for popularizing the matrix formulation of the LU -decomposition is often given to the British mathematician Alan Turing for his work on the subject in 1948. Turing, one of the great geniuses of the twentieth century, is the founder of the field of artificial intelligence. Among his many accomplishments in that field, he developed the concept of an internally programmed computer before the practical technology had reached the point where the construction of such a machine was possible. During World War II Turing was secretly recruited by the British government’s Code and Cypher School at Bletchley Park to help break the Nazi Enigma codes; it was Turing’s statistical approach that provided the breakthrough. In addition to being a brilliant mathematician, Turing was a world-class runner who competed successfully with Olympic-level competition. Sadly, Turing, a homosexual, was tried and convicted of “gross indecency” in 1952, in violation of the then-existing British statutes. Depressed, he committed suicide at age 41 by eating an apple laced with cyanide.

[Image: Science Source/Science Source]

Not every square matrix has an LU -decomposition. However, if it is possible to reduce a square matrix A to row echelon form by Gaussian elimination *without performing any row interchanges*, then A will have an LU -decomposition, though it may not be unique. To see why this is so, assume that A has been reduced to a row echelon form U using a sequence of row operations that does not include row interchanges. We know from Theorem 1.5.1 that these operations can be accomplished by multiplying A on the left by an appropriate sequence of elementary matrices; that is, there exist elementary matrices $E_1, E_2, \dots, E_k$ such that

$$E_k \cdots E_2 E_1 A = U \quad (8)$$

Since elementary matrices are invertible, we can solve (8) for A as

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} U$$

or more briefly as

$$A = LU \quad (9)$$

where

$$L = E_1^{-1} E_2^{-1} \cdots E_k^{-1} \quad (10)$$

We now have all of the ingredients to prove the following result.

Theorem 9.1.1

If A is a square matrix that can be reduced to a row echelon form U by Gaussian elimination without row interchanges, then A can be factored as $A = LU$, where L is a lower triangular matrix.

Proof Let L and U be the matrices in Formulas (10) and (8), respectively. The matrix U is upper triangular because it is a row echelon form of a square matrix (so all entries below its main diagonal are zero). To prove that L is lower triangular, it suffices to prove that each factor on the right side of (10) is lower triangular, since Theorem 1.7.1(b) will then imply that L itself is lower triangular. Since row interchanges are excluded, each E_j results either by adding a scalar multiple of one row of an identity matrix to a row below or by multiplying one row of an identity matrix by a nonzero scalar. In either case, the resulting matrix E_j is lower triangular and hence so is E_j^{-1} by Theorem 1.7.1(d). This completes the proof. ■

EXAMPLE 2 | An LU -Decomposition

Find an LU -decomposition of

$$A = \begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix}$$

Solution To obtain an LU -decomposition, $A = LU$, we will reduce A to a row echelon form U using Gaussian elimination and then calculate L from (10). The steps are as follows:

	Reduction to Row Echelon Form	Row Operation	Elementary Matrix Corresponding to the Row Operation	Inverse of the Elementary Matrix
	$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix}$			
Step 1.	$\begin{bmatrix} 1 & 3 & 1 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix}$	$\frac{1}{2} \times \text{row 1}$	$E_1 = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$E_1^{-1} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Step 2.	$\begin{bmatrix} 1 & 3 & 1 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix}$	$(3 \times \text{row 1}) + \text{row 2}$	$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Step 3.	$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 4 & 9 & 2 \end{bmatrix}$	$(-4 \times \text{row 1}) + \text{row 3}$	$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$	$E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$
Step 4.	$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & -3 & -2 \end{bmatrix}$	$(3 \times \text{row 2}) + \text{row 3}$	$E_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$	$E_4^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$
Step 5.	$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 7 \end{bmatrix}$	$\frac{1}{7} \times \text{row 3}$	$E_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{7} \end{bmatrix}$	$E_5^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix}$
	$\begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} = U$			
and, from (10),				
$L = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix}$ $= \begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \quad (11)$				
so				
$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$				
is an LU -decomposition of A .				

Bookkeeping

As Example 2 shows, most of the work in constructing an LU -decomposition is expended in calculating L . However, *all* this work can be eliminated by some careful bookkeeping of the operations used to reduce A to U .

Because we are assuming that no row interchanges are required to reduce A to U , there are only two types of operations involved—multiplying a row by a nonzero constant, and adding a scalar multiple of one row to another. The first operation is used to introduce the leading 1's and the second to introduce zeros below the leading 1's.

In Example 2, a multiplier of $\frac{1}{2}$ was needed in Step 1 to introduce a leading 1 in the first row, and a multiplier of $\frac{1}{7}$ was needed in Step 5 to introduce a leading 1 in the third row. No actual multiplier was required to introduce a leading 1 in the second row because it was already a 1 at the end of Step 2, but for convenience let us say that the multiplier was 1. Comparing these multipliers with the successive diagonal entries of L , we see that these diagonal entries are precisely the reciprocals of the multipliers used to construct U :

$$L = \begin{bmatrix} \textcircled{2} & 0 & 0 \\ -3 & \textcircled{1} & 0 \\ 4 & -3 & \textcircled{7} \end{bmatrix} \quad (12)$$

Also observe in Example 2 that to introduce zeros below the leading 1 in the first row, we used the operations

add 3 times the first row to the second

add -4 times the first row to the third

and to introduce the zero below the leading 1 in the second row, we used the operation

add 3 times the second row to the third

Now note in (11) that in each position below the main diagonal of L , the entry is the *negative* of the multiplier in the operation that introduced the zero in that position in U . This suggests the following procedure for constructing an LU -decomposition of a square matrix A , assuming that this matrix can be reduced to row echelon form without row interchanges.

Procedure for Constructing an LU -Decomposition

Step 1. Reduce A to a row echelon form U by Gaussian elimination without row interchanges, keeping track of the multipliers used to introduce the leading 1's and the multipliers used to introduce the zeros below the leading 1's.

Step 2. In each position along the main diagonal of L , place the reciprocal of the multiplier that introduced the leading 1 in that position in U .

Step 3. In each position below the main diagonal of L , place the negative of the multiplier used to introduce the zero in that position in U .

Step 4. Form the decomposition $A = LU$.

EXAMPLE 3 | Constructing an LU -Decomposition

Find an LU -decomposition of

$$A = \begin{bmatrix} 6 & -2 & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix}$$

Solution We will reduce A to a row echelon form U and at each step we will fill in an entry of L in accordance with the four-step procedure above.

$$\begin{array}{lcl}
 A = \begin{bmatrix} 6 & -2 & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix} & \begin{bmatrix} \bullet & 0 & 0 \\ \bullet & \bullet & 0 \\ \bullet & \bullet & \bullet \end{bmatrix} & \begin{array}{l} \bullet \text{ denotes an unknown} \\ \text{entry of } L. \end{array} \\
 \\
 \begin{bmatrix} \textcircled{1} & -\frac{1}{3} & 0 \\ 9 & -1 & 1 \\ 3 & 7 & 5 \end{bmatrix} \leftarrow \text{multiplier} = \frac{1}{6} & \begin{bmatrix} 6 & 0 & 0 \\ \bullet & \bullet & 0 \\ \bullet & \bullet & \bullet \end{bmatrix} & \\
 \\
 \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ \textcircled{0} & 2 & 1 \\ \textcircled{0} & 8 & 5 \end{bmatrix} \begin{array}{l} \leftarrow \text{multiplier} = -9 \\ \leftarrow \text{multiplier} = -3 \end{array} & \begin{bmatrix} 6 & 0 & 0 \\ 9 & \bullet & 0 \\ 3 & \bullet & \bullet \end{bmatrix} & \\
 \\
 \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & \textcircled{1} & \frac{1}{2} \\ 0 & 8 & 5 \end{bmatrix} \leftarrow \text{multiplier} = \frac{1}{2} & \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & \bullet & \bullet \end{bmatrix} & \\
 \\
 \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & \textcircled{0} & 1 \end{bmatrix} \leftarrow \text{multiplier} = -8 & \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & \bullet \end{bmatrix} & \\
 \\
 U = \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & \textcircled{1} \end{bmatrix} \leftarrow \text{multiplier} = 1 & L = \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & 1 \end{bmatrix} & \begin{array}{l} \text{No actual operation is} \\ \text{performed here since} \\ \text{there is already a leading} \\ \text{1 in the third row.} \end{array}
 \end{array}$$

Thus, we have constructed the LU -decomposition

$$A = LU = \begin{bmatrix} 6 & 0 & 0 \\ 9 & 2 & 0 \\ 3 & 8 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{1}{3} & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}$$

We leave it for you to confirm this end result by multiplying the factors.

LU -Decompositions Are Not Unique

In general, LU -decompositions are not unique. For example, if

$$A = LU = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

and L has nonzero diagonal entries (which will be true if A is invertible), then we can shift the diagonal entries from the left factor to the right factor by writing

$$\begin{aligned}
 A &= \begin{bmatrix} 1 & 0 & 0 \\ l_{21}/l_{11} & 1 & 0 \\ l_{31}/l_{11} & l_{32}/l_{22} & 1 \end{bmatrix} \begin{bmatrix} l_{11} & 0 & 0 \\ 0 & l_{22} & 0 \\ 0 & 0 & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ l_{21}/l_{11} & 1 & 0 \\ l_{31}/l_{11} & l_{32}/l_{22} & 1 \end{bmatrix} \begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ 0 & l_{22} & l_{22}u_{23} \\ 0 & 0 & l_{33} \end{bmatrix}
 \end{aligned}$$

which is another LU -decomposition of A .

LDU-Decompositions

The method we have given for computing LU -decompositions may result in an “asymmetry” in that the matrix U has 1’s on the main diagonal but L need not. However, if it is preferred to have 1’s on the main diagonal of both the lower triangular factor and the upper triangular factor, then we can “shift” the diagonal entries of L to a diagonal matrix D and write L as

$$L = L'D$$

where L' is a lower triangular matrix with 1’s on the main diagonal. For example, a general 3×3 lower triangular matrix with nonzero entries on the main diagonal can be factored as

$$\begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ a_{21}/a_{11} & 1 & 0 \\ a_{31}/a_{11} & a_{32}/a_{22} & 1 \end{bmatrix} \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$$

$L \qquad \qquad L' \qquad \qquad D$

Note that the columns of L' are obtained by dividing each entry in the corresponding column of L by the diagonal entry in the column. Thus, for example, we can rewrite (4) as

$$\begin{bmatrix} 2 & 6 & 2 \\ -3 & -8 & 0 \\ 4 & 9 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -3 & 7 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ 2 & -3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix} \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \quad (13)$$

One can prove that if A is an invertible matrix that can be reduced to row echelon form without row interchanges, then A can be factored *uniquely* as

$$A = LDU$$

where L is a lower triangular matrix with 1’s on the main diagonal, D is a diagonal matrix, and U is an upper triangular matrix with 1’s on the main diagonal. This is called the **LDU-decomposition** (or **LDU-factorization**) of A .

If desired, the diagonal matrix and the upper triangular matrix in (13) can be multiplied to produce an LU -decomposition in which the 1’s are on the main diagonal of L rather than U .

PLU-Decompositions

Many computer algorithms for solving linear systems perform row interchanges to reduce roundoff error, in which case the existence of an LU -decomposition is not guaranteed. However, it is possible to work around this problem by “preprocessing” the coefficient matrix A so that the row interchanges are performed *prior* to computing the LU -decomposition itself. The idea is to create a matrix Q (called a **permutation matrix**) by multiplying, in sequence, those elementary matrices that produce the row interchanges and then execute them by computing the product QA . This product can then be reduced to row echelon form *without* row interchanges, so it is assured to have an LU -decomposition

$$QA = LU \quad (14)$$

Because the matrix Q is invertible (being a product of elementary matrices), the systems $A\mathbf{x} = \mathbf{b}$ and $QA\mathbf{x} = Q\mathbf{b}$ will have the same solutions. But it follows from (14) that the latter system can be rewritten as $LU\mathbf{x} = Q\mathbf{b}$ and hence can be solved using LU -decomposition.

It is common to see Equation (14) expressed as

$$A = PLU \quad (15)$$

in which $P = Q^{-1}$. This is called a **PLU-decomposition** or (**PLU-factorization**) of A .

Exercise Set 9.1

1. Use the method of Example 1 and the
- LU
- decomposition

$$\begin{bmatrix} 3 & -6 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

to solve the system

$$\begin{aligned} 3x_1 - 6x_2 &= 0 \\ -2x_1 + 5x_2 &= 1 \end{aligned}$$

2. Use the method of Example 1 and the
- LU
- decomposition

$$\begin{bmatrix} 3 & -6 & -3 \\ 2 & 0 & 6 \\ -4 & 7 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 4 & 0 \\ -4 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

to solve the system

$$\begin{aligned} 3x_1 - 6x_2 - 3x_3 &= -3 \\ 2x_1 &+ 6x_3 = -22 \\ -4x_1 + 7x_2 + 4x_3 &= 3 \end{aligned}$$

In Exercises 3–6, find an LU -decomposition of the coefficient matrix, and then use the method of Example 1 to solve the system.

$$3. \begin{bmatrix} 2 & 8 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

$$4. \begin{bmatrix} -5 & -10 \\ 6 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -10 \\ 19 \end{bmatrix}$$

$$5. \begin{bmatrix} 2 & -2 & -2 \\ 0 & -2 & 2 \\ -1 & 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \\ 6 \end{bmatrix}$$

$$6. \begin{bmatrix} -3 & 12 & -6 \\ 1 & -2 & 2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -33 \\ 7 \\ -1 \end{bmatrix}$$

In Exercises 7–8, an LU -decomposition of a matrix A is given.

- a. Compute
- L^{-1}
- and
- U^{-1}
- .

- b. Use the result in part (a) to find the inverse of
- A
- .

$$7. A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 1 \\ -6 & -1 & 2 \end{bmatrix};$$

$$A = LU = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & -5 \\ 0 & 0 & 6 \end{bmatrix}$$

8. The
- LU
- decomposition obtained in Example 2.

9. Let

$$A = \begin{bmatrix} 2 & 1 & -1 \\ -2 & -1 & 2 \\ 2 & 1 & 0 \end{bmatrix}$$

- a. Find an
- LU
- decomposition of
- A
- .

- b. Express
- A
- in the form
- $A = L_1 D U_1$
- , where
- L_1
- is lower triangular with 1's along the main diagonal,
- U_1
- is upper triangular, and
- D
- is a diagonal matrix.

- c. Express
- A
- in the form
- $A = L_2 U_2$
- , where
- L_2
- is lower triangular with 1's along the main diagonal and
- U_2
- is upper triangular.

10. a. Show that the matrix

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

has no LU -decomposition.

- b. Find a
- PLU
- decomposition of this matrix.

In Exercises 11–12, use the given PLU -decomposition of A to solve the linear system $A\mathbf{x} = \mathbf{b}$ by rewriting it as $P^{-1}A\mathbf{x} = P^{-1}\mathbf{b}$ and solving this system by LU -decomposition.

$$11. \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}; A = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 2 & 2 \\ 3 & 1 & 3 \end{bmatrix};$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & -5 & 17 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} = PLU$$

$$12. \mathbf{b} = \begin{bmatrix} 3 \\ 0 \\ 6 \end{bmatrix}; A = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 2 & 1 \\ 8 & 1 & 8 \end{bmatrix};$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 4 & 1 & 2 \\ 0 & -1 & 4 \\ 0 & 0 & 9 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{4} & \frac{1}{2} \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix} = PLU$$

In Exercises 13–14, find the LDU -decomposition of A .

$$13. A = \begin{bmatrix} 2 & 2 \\ 4 & 1 \end{bmatrix} \quad 14. A = \begin{bmatrix} 3 & -12 & 6 \\ 0 & 2 & 0 \\ 6 & -28 & 13 \end{bmatrix}$$

In Exercises 15–16, find a PLU -decomposition of A , and use it to solve the linear system $A\mathbf{x} = \mathbf{b}$ by the method of Exercises 11 and 12.

$$15. A = \begin{bmatrix} 3 & -1 & 0 \\ 3 & -1 & 1 \\ 0 & 2 & 1 \end{bmatrix}; \mathbf{b} = \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}$$

$$16. A = \begin{bmatrix} 0 & 3 & -2 \\ 1 & 1 & 4 \\ 2 & 2 & 5 \end{bmatrix}; \mathbf{b} = \begin{bmatrix} 7 \\ 5 \\ -2 \end{bmatrix}$$

17. Let $A\mathbf{x} = \mathbf{b}$ be a linear system of n equations in n unknowns, and assume that A is an invertible matrix that can be reduced to row echelon form without row interchanges. How many additions and multiplications are required to solve the system by the method of Example 1?

Working with Proofs

18. Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

- a. Prove: If $a \neq 0$, then the matrix A has a unique LU -decomposition with 1's along the main diagonal of L .
- b. Find the LU -decomposition described in part (a).

19. Prove: If A is any $n \times n$ matrix, then A can be factored as $A = PLU$, where L is lower triangular, U is upper triangular, and P can be obtained by interchanging the rows of I_n appropriately. [Hint: Let U be a row echelon form of A , and let all row interchanges required in the reduction of A to U be performed first.]

True-False Exercises

- TF. In parts (a)–(e) determine whether the statement is true or false, and justify your answer.
- Every square matrix has an LU -decomposition.
 - If a square matrix A is row equivalent to an upper triangular matrix U , then A has an LU -decomposition.
 - If $L_1, L_2, \dots, L_k$ are $n \times n$ lower triangular matrices, then the product $L_1 L_2 \cdots L_k$ is lower triangular.
 - If an invertible matrix A has an LU -decomposition, then A has a unique LDU -decomposition.
 - Every square matrix has a PLU -decomposition.

Working with Technology

- T1. Technology utilities vary in how they handle LU -decompositions. For example, many utilities perform row interchanges to reduce roundoff error and hence produce PLU -decompositions, even when asked for LU -decompositions. See what happens when you use your utility to find an LU -decomposition of the matrix A in Example 2.

- T2. The accompanying figure shows a metal plate whose edges are held at the temperatures shown. It follows from thermodynamic principles that the temperature at each of the six interior nodes will eventually stabilize at a value that is approximately the average of the temperatures at the four neighboring nodes. These are called the **steady-state temperatures** at the nodes. Thus, for example, if we denote the steady-state temperatures at the interior nodes in the accompanying figure as T_1, T_2, T_3, T_4, T_5 , and T_6 , then at the node labeled T_1 that temperature will be $T_1 = \frac{1}{4}(0 + 5 + T_2 + T_3)$ or, equivalently,

$$4T_1 - T_2 - T_3 = 5$$

Find a linear system whose solution gives the steady-state temperatures at the nodes, and use your technology utility to solve that system by LU -decomposition.

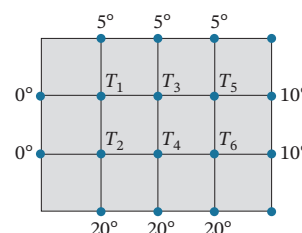


FIGURE Ex-T2

9.2 The Power Method

The eigenvalues of a square matrix can, in theory, be found by solving the characteristic equation. However, this procedure has so many computational difficulties that it is almost never used in applications. In this section we will discuss an algorithm that can be used to approximate the eigenvalue with greatest absolute value and a corresponding eigenvector. This particular eigenvalue and its corresponding eigenvectors are important because they arise naturally in many iterative processes. The methods we will study in this section have recently been used to create Internet search engines such as Google.

The Power Method

There are many applications in which some vector $\mathbf{x}_0$ in R^n is multiplied repeatedly by an $n \times n$ matrix A to produce a sequence

$$\mathbf{x}_0, \quad A\mathbf{x}_0, \quad A^2\mathbf{x}_0, \dots, \quad A^k\mathbf{x}_0, \dots$$

We call a sequence of this form a **power sequence generated by A** . In this section we will be concerned with the convergence of power sequences and how such sequences can be used to approximate eigenvalues and eigenvectors. For this purpose, we make the following definition.

Definition 1

If the *distinct* eigenvalues of a matrix A are $\lambda_1, \lambda_2, \dots, \lambda_k$, and if $|\lambda_1|$ is larger than $|\lambda_2|, \dots, |\lambda_k|$, then λ_1 is called a **dominant eigenvalue** of A . Any eigenvector corresponding to a dominant eigenvalue is called a **dominant eigenvector** of A .

EXAMPLE 1 | Dominant Eigenvalues

Some matrices have dominant eigenvalues and some do not. For example, if the distinct eigenvalues of a matrix are

$$\lambda_1 = -4, \quad \lambda_2 = -2, \quad \lambda_3 = 1, \quad \lambda_4 = 3$$

then $\lambda_1 = -4$ is dominant since $|\lambda_1| = 4$ is greater than the absolute values of all the other eigenvalues; but if the distinct eigenvalues of a matrix are

$$\lambda_1 = 7, \quad \lambda_2 = -7, \quad \lambda_3 = -2, \quad \lambda_4 = 5$$

then $|\lambda_1| = |\lambda_2| = 7$, so there is no single eigenvalue whose absolute value is greater than the absolute value of all the other eigenvalues.

The most important theorems about convergence of power sequences apply to $n \times n$ matrices with n linearly independent eigenvectors (symmetric matrices, for example), so we will limit our discussion to this case in this section.

Theorem 9.2.1

Let A be a symmetric $n \times n$ matrix that has a positive* dominant eigenvalue λ . If $\mathbf{x}_0$ is a unit vector in R^n that is not orthogonal to the eigenspace corresponding to λ , then the normalized power sequence

$$\mathbf{x}_0, \quad \mathbf{x}_1 = \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|}, \quad \mathbf{x}_2 = \frac{A\mathbf{x}_1}{\|A\mathbf{x}_1\|}, \dots, \quad \mathbf{x}_k = \frac{A\mathbf{x}_{k-1}}{\|A\mathbf{x}_{k-1}\|}, \dots \quad (1)$$

converges to a unit dominant eigenvector, and the sequence

$$A\mathbf{x}_0 \cdot \mathbf{x}_0, \quad A\mathbf{x}_1 \cdot \mathbf{x}_1, \dots, \quad A\mathbf{x}_k \cdot \mathbf{x}_k, \dots \quad (2)$$

converges to the dominant eigenvalue λ .

Remark In the exercises we will ask you to show that (1) can also be expressed as

$$\mathbf{x}_0, \quad \mathbf{x}_1 = \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|}, \quad \mathbf{x}_2 = \frac{A^2\mathbf{x}_0}{\|A^2\mathbf{x}_0\|}, \dots, \quad \mathbf{x}_k = \frac{A^k\mathbf{x}_0}{\|A^k\mathbf{x}_0\|}, \dots \quad (3)$$

This form of the power sequence expresses each iterate in terms of the starting vector $\mathbf{x}_0$, rather than in terms of its predecessor.

We will not prove Theorem 9.2.1, but we can make it plausible geometrically in the 2×2 case where A is a symmetric matrix with distinct positive eigenvalues, λ_1 and λ_2 , one of which is dominant. To be specific, assume that λ_1 is dominant and

$$\lambda_1 > \lambda_2 > 0$$

*If the dominant eigenvalue is not positive, sequence (2) will still converge to the dominant eigenvalue, but sequence (1) may not converge to a *specific* dominant eigenvector because of *alternation* (see Exercise 11). Nevertheless, each term of (1) will closely approximate *some* dominant eigenvector for sufficiently large values of k .

Since we are assuming that A is symmetric and has distinct eigenvalues, it follows from Theorem 7.2.2 that the eigenspaces corresponding to λ_1 and λ_2 are perpendicular lines through the origin. Thus, the assumption that $\mathbf{x}_0$ is a unit vector that is not orthogonal to the eigenspace corresponding to λ_1 implies that $\mathbf{x}_0$ does not lie in the eigenspace corresponding to λ_2 . To see the geometric effect of multiplying $\mathbf{x}_0$ by A , it will be useful to split $\mathbf{x}_0$ into the sum

$$\mathbf{x}_0 = \mathbf{v}_0 + \mathbf{w}_0 \quad (4)$$

where $\mathbf{v}_0$ and $\mathbf{w}_0$ are the orthogonal projections of $\mathbf{x}_0$ on the eigenspaces of λ_1 and λ_2 , respectively (Figure 9.2.1a).

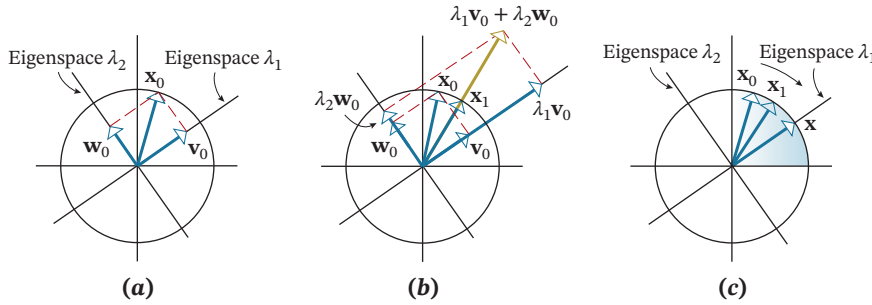


FIGURE 9.2.1

This enables us to express $A\mathbf{x}_0$ as

$$A\mathbf{x}_0 = A\mathbf{v}_0 + A\mathbf{w}_0 = \lambda_1\mathbf{v}_0 + \lambda_2\mathbf{w}_0 \quad (5)$$

which tells us that multiplying $\mathbf{x}_0$ by A “scales” the terms $\mathbf{v}_0$ and $\mathbf{w}_0$ in (4) by λ_1 and λ_2 , respectively. However, λ_1 is larger than λ_2 , so the scaling is greater in the direction of $\mathbf{v}_0$ than in the direction of $\mathbf{w}_0$. Thus, multiplying $\mathbf{x}_0$ by A “pulls” $\mathbf{x}_0$ toward the eigenspace of λ_1 , and normalizing produces a vector $\mathbf{x}_1 = A\mathbf{x}_0 / \|A\mathbf{x}_0\|$, which is on the unit circle and is closer to the eigenspace of λ_1 than $\mathbf{x}_0$ (Figure 9.2.1b). Similarly, multiplying $\mathbf{x}_1$ by A and normalizing produces a unit vector $\mathbf{x}_2$ that is closer to the eigenspace of λ_1 than $\mathbf{x}_1$. Thus, it seems reasonable that by repeatedly multiplying by A and normalizing we will produce a sequence of vectors $\mathbf{x}_k$ that lie on the unit circle and converge to a unit vector $\mathbf{x}$ in the eigenspace of λ_1 (Figure 9.2.1c). Moreover, if $\mathbf{x}_k$ converges to $\mathbf{x}$, then it also seems reasonable that $A\mathbf{x}_k \cdot \mathbf{x}_k$ will converge to

$$A\mathbf{x} \cdot \mathbf{x} = \lambda_1 \mathbf{x} \cdot \mathbf{x} = \lambda_1 \|\mathbf{x}\|^2 = \lambda_1$$

which is the dominant eigenvalue of A .

The Power Method with Euclidean Scaling

Theorem 9.2.1 provides us with an algorithm for approximating the dominant eigenvalue and a corresponding unit eigenvector of a symmetric matrix A , provided the dominant eigenvalue is positive. This algorithm, called the **power method with Euclidean scaling**, is as follows:

The Power Method with Euclidean Scaling

- Step 0.** Choose an arbitrary nonzero vector and normalize it, if need be, to obtain a unit vector $\mathbf{x}_0$.
- Step 1.** Compute $A\mathbf{x}_0$ and normalize it to obtain the first approximation $\mathbf{x}_1$ to a dominant unit eigenvector. Compute $A\mathbf{x}_1 \cdot \mathbf{x}_1$ to obtain the first approximation to the dominant eigenvalue.

Step 2. Compute $A\mathbf{x}_1$ and normalize it to obtain the second approximation $\mathbf{x}_2$ to a dominant unit eigenvector. Compute $A\mathbf{x}_2 \cdot \mathbf{x}_2$ to obtain the second approximation to the dominant eigenvalue.

Step 3. Compute $A\mathbf{x}_2$ and normalize it to obtain the third approximation $\mathbf{x}_3$ to a dominant unit eigenvector. Compute $A\mathbf{x}_3 \cdot \mathbf{x}_3$ to obtain the third approximation to the dominant eigenvalue.

Continuing in this way will usually generate a sequence of better and better approximations to the dominant eigenvalue and a corresponding unit eigenvector.*

EXAMPLE 2 | The Power Method with Euclidean Scaling

Apply the power method with Euclidean scaling to

$$A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \quad \text{with} \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Stop at $\mathbf{x}_5$ and compare the resulting approximations to the exact values of the dominant eigenvalue and eigenvector.

Solution We will leave it for you to show that the eigenvalues of A are $\lambda = 1$ and $\lambda = 5$ and that the eigenspace corresponding to the dominant eigenvalue $\lambda = 5$ is the line represented by the parametric equations $x_1 = t, x_2 = t$, which we can write in vector form as

$$\mathbf{x} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (6)$$

Setting $t = 1/\sqrt{2}$ yields the normalized dominant eigenvector

$$\mathbf{v}_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \approx \begin{bmatrix} 0.707106781187\dots \\ 0.707106781187\dots \end{bmatrix} \quad (7)$$

Now let us see what happens when we use the power method, starting with the unit vector $\mathbf{x}_0$.

$$A\mathbf{x}_0 = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \mathbf{x}_1 = \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|} = \frac{1}{\sqrt{13}} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \approx \frac{1}{3.60555} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \approx \begin{bmatrix} 0.83205 \\ 0.55470 \end{bmatrix}$$

$$A\mathbf{x}_1 \approx \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 0.83205 \\ 0.55470 \end{bmatrix} \approx \begin{bmatrix} 3.60555 \\ 3.32820 \end{bmatrix} \quad \mathbf{x}_2 = \frac{A\mathbf{x}_1}{\|A\mathbf{x}_1\|} \approx \frac{1}{4.90682} \begin{bmatrix} 3.60555 \\ 3.32820 \end{bmatrix} \approx \begin{bmatrix} 0.73480 \\ 0.67828 \end{bmatrix}$$

$$A\mathbf{x}_2 \approx \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 0.73480 \\ 0.67828 \end{bmatrix} \approx \begin{bmatrix} 3.56097 \\ 3.50445 \end{bmatrix} \quad \mathbf{x}_3 = \frac{A\mathbf{x}_2}{\|A\mathbf{x}_2\|} \approx \frac{1}{4.99616} \begin{bmatrix} 3.56097 \\ 3.50445 \end{bmatrix} \approx \begin{bmatrix} 0.71274 \\ 0.70143 \end{bmatrix}$$

$$A\mathbf{x}_3 \approx \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 0.71274 \\ 0.70143 \end{bmatrix} \approx \begin{bmatrix} 3.54108 \\ 3.52976 \end{bmatrix} \quad \mathbf{x}_4 = \frac{A\mathbf{x}_3}{\|A\mathbf{x}_3\|} \approx \frac{1}{4.99985} \begin{bmatrix} 3.54108 \\ 3.52976 \end{bmatrix} \approx \begin{bmatrix} 0.70824 \\ 0.70597 \end{bmatrix}$$

$$A\mathbf{x}_4 \approx \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 0.70824 \\ 0.70597 \end{bmatrix} \approx \begin{bmatrix} 3.53666 \\ 3.53440 \end{bmatrix} \quad \mathbf{x}_5 = \frac{A\mathbf{x}_4}{\|A\mathbf{x}_4\|} \approx \frac{1}{4.99999} \begin{bmatrix} 3.53666 \\ 3.53440 \end{bmatrix} \approx \begin{bmatrix} 0.70733 \\ 0.70688 \end{bmatrix}$$

*If the vector $\mathbf{x}_0$ happens to be orthogonal to the eigenspace of the dominant eigenvalue, then the hypotheses of Theorem 9.2.1 will be violated and the method may fail. However, the reality is that computer roundoff errors usually perturb $\mathbf{x}_0$ enough to destroy any orthogonality and make the algorithm work. This is one instance in which errors help to obtain correct results!

$$\lambda^{(1)} = (A\mathbf{x}_1) \cdot \mathbf{x}_1 = (A\mathbf{x}_1)^T \mathbf{x}_1 \approx \begin{bmatrix} 3.60555 & 3.32820 \end{bmatrix} \begin{bmatrix} 0.83205 \\ 0.55470 \end{bmatrix} \approx 4.84615$$

$$\lambda^{(2)} = (A\mathbf{x}_2) \cdot \mathbf{x}_2 = (A\mathbf{x}_2)^T \mathbf{x}_2 \approx \begin{bmatrix} 3.56097 & 3.50445 \end{bmatrix} \begin{bmatrix} 0.73480 \\ 0.67828 \end{bmatrix} \approx 4.99361$$

$$\lambda^{(3)} = (A\mathbf{x}_3) \cdot \mathbf{x}_3 = (A\mathbf{x}_3)^T \mathbf{x}_3 \approx \begin{bmatrix} 3.54108 & 3.52976 \end{bmatrix} \begin{bmatrix} 0.71274 \\ 0.70143 \end{bmatrix} \approx 4.99974$$

$$\lambda^{(4)} = (A\mathbf{x}_4) \cdot \mathbf{x}_4 = (A\mathbf{x}_4)^T \mathbf{x}_4 \approx \begin{bmatrix} 3.53666 & 3.53440 \end{bmatrix} \begin{bmatrix} 0.70824 \\ 0.70597 \end{bmatrix} \approx 4.99999$$

$$\lambda^{(5)} = (A\mathbf{x}_5) \cdot \mathbf{x}_5 = (A\mathbf{x}_5)^T \mathbf{x}_5 \approx \begin{bmatrix} 3.53576 & 3.53531 \end{bmatrix} \begin{bmatrix} 0.70733 \\ 0.70688 \end{bmatrix} \approx 5.00000$$

Thus, $\lambda^{(5)}$ approximates the dominant eigenvalue to five decimal place accuracy and $\mathbf{x}_5$ approximates the dominant eigenvector in (7) to three decimal place accuracy.

It is accidental that $\lambda^{(5)}$ (the fifth approximation) produced five decimal place accuracy. In general, n iterations need not produce n decimal place accuracy.

The Power Method with Maximum Entry Scaling

There is a variation of the power method in which the iterates, rather than being normalized at each stage, are scaled to make the maximum entry 1. To describe this method, it will be convenient to denote the maximum *absolute value* of the entries in a vector $\mathbf{x}$ by $\max(\mathbf{x})$. Thus, for example, if

$$\mathbf{x} = \begin{bmatrix} 5 \\ 3 \\ -7 \\ 2 \end{bmatrix}$$

then $\max(\mathbf{x}) = 7$. We will need the following variation of Theorem 9.2.1.

Theorem 9.2.2

Let A be a symmetric $n \times n$ matrix that has a positive dominant* eigenvalue λ . If $\mathbf{x}_0$ is a nonzero vector in R^n that is not orthogonal to the eigenspace corresponding to λ , then the sequence

$$\mathbf{x}_0, \quad \mathbf{x}_1 = \frac{A\mathbf{x}_0}{\max(A\mathbf{x}_0)}, \quad \mathbf{x}_2 = \frac{A\mathbf{x}_1}{\max(A\mathbf{x}_1)}, \dots, \quad \mathbf{x}_k = \frac{A\mathbf{x}_{k-1}}{\max(A\mathbf{x}_{k-1})}, \dots \quad (8)$$

converges to an eigenvector corresponding to λ , and the sequence

$$\frac{A\mathbf{x}_0 \cdot \mathbf{x}_0}{\mathbf{x}_0 \cdot \mathbf{x}_0}, \quad \frac{A\mathbf{x}_1 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}, \dots, \quad \frac{A\mathbf{x}_k \cdot \mathbf{x}_k}{\mathbf{x}_k \cdot \mathbf{x}_k}, \dots \quad (9)$$

converges to λ .

Remark In the exercises we will ask you to show that (8) can be written in the alternative form

$$\mathbf{x}_0, \quad \mathbf{x}_1 = \frac{A\mathbf{x}_0}{\max(A\mathbf{x}_0)}, \quad \mathbf{x}_2 = \frac{A^2\mathbf{x}_0}{\max(A^2\mathbf{x}_0)}, \dots, \quad \mathbf{x}_k = \frac{A^k\mathbf{x}_0}{\max(A^k\mathbf{x}_0)}, \dots \quad (10)$$

which expresses the iterates in terms of the initial vector $\mathbf{x}_0$.

*As in Theorem 9.2.1, if the dominant eigenvalue is not positive, sequence (9) will still converge to the dominant eigenvalue, but sequence (8) may not converge to a *specific* dominant eigenvector. Nevertheless, each term of (8) will closely approximate *some* dominant eigenvector for sufficiently large values of k (see Exercise 11).

We will omit the proof of this theorem, but if we accept that (8) converges to an eigenvector of A , then it is not hard to see why (9) converges to the dominant eigenvalue. To see this, note that each term in (9) is of the form

$$\frac{A\mathbf{x} \cdot \mathbf{x}}{\mathbf{x} \cdot \mathbf{x}} \quad (11)$$

which is called a **Rayleigh quotient** of A . In the case where λ is an eigenvalue of A and $\mathbf{x}$ is a corresponding eigenvector, the Rayleigh quotient is

$$\frac{A\mathbf{x} \cdot \mathbf{x}}{\mathbf{x} \cdot \mathbf{x}} = \frac{\lambda\mathbf{x} \cdot \mathbf{x}}{\mathbf{x} \cdot \mathbf{x}} = \frac{\lambda(\mathbf{x} \cdot \mathbf{x})}{\mathbf{x} \cdot \mathbf{x}} = \lambda$$

Thus, if $\mathbf{x}_k$ converges to a dominant eigenvector $\mathbf{x}$, then it seems reasonable that

$$\frac{A\mathbf{x}_k \cdot \mathbf{x}_k}{\mathbf{x}_k \cdot \mathbf{x}_k} \text{ converges to } \frac{A\mathbf{x} \cdot \mathbf{x}}{\mathbf{x} \cdot \mathbf{x}} = \lambda$$

which is the dominant eigenvalue.

Theorem 9.2.2 produces the following algorithm, which is called the **power method with maximum entry scaling**.

The Power Method with Maximum Entry Scaling

Step 0. Choose an arbitrary nonzero vector $\mathbf{x}_0$.

Step 1. Compute $A\mathbf{x}_0$ and multiply it by the factor $1/\max(A\mathbf{x}_0)$ to obtain the first approximation $\mathbf{x}_1$ to a dominant eigenvector. Compute the Rayleigh quotient of $\mathbf{x}_1$ to obtain the first approximation to the dominant eigenvalue.

Step 2. Compute $A\mathbf{x}_1$ and scale it by the factor $1/\max(A\mathbf{x}_1)$ to obtain the second approximation $\mathbf{x}_2$ to a dominant eigenvector. Compute the Rayleigh quotient of $\mathbf{x}_2$ to obtain the second approximation to the dominant eigenvalue.

Step 3. Compute $A\mathbf{x}_2$ and scale it by the factor $1/\max(A\mathbf{x}_2)$ to obtain the third approximation $\mathbf{x}_3$ to a dominant eigenvector. Compute the Rayleigh quotient of $\mathbf{x}_3$ to obtain the third approximation to the dominant eigenvalue.

Continuing in this way will generate a sequence of better and better approximations to the dominant eigenvalue and a corresponding eigenvector.

Historical Note



The British mathematical physicist John Rayleigh won the Nobel prize in physics in 1904 for his discovery of the inert gas argon. Rayleigh also made fundamental discoveries in acoustics and optics, and his work in wave phenomena enabled him to give the first accurate explanation of why the sky is blue.

[Image: The Granger Collection, New York]

**John William
Strutt Rayleigh**
(1842–1919)

EXAMPLE 3 | Example 2 Revisited Using Maximum Entry Scaling

Apply the power method with maximum entry scaling to

$$A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \quad \text{with} \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Stop at $\mathbf{x}_5$ and compare the resulting approximations to the exact values and to the approximations obtained in Example 2.

Solution We leave it for you to confirm that

$$\begin{aligned} A\mathbf{x}_0 &= \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} & \mathbf{x}_1 &= \frac{A\mathbf{x}_0}{\max(A\mathbf{x}_0)} = \frac{1}{3} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \approx \begin{bmatrix} 1.00000 \\ 0.66667 \end{bmatrix} \\ A\mathbf{x}_1 &\approx \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1.00000 \\ 0.66667 \end{bmatrix} \approx \begin{bmatrix} 4.33333 \\ 4.00000 \end{bmatrix} & \mathbf{x}_2 &= \frac{A\mathbf{x}_1}{\max(A\mathbf{x}_1)} \approx \frac{1}{4.33333} \begin{bmatrix} 4.33333 \\ 4.00000 \end{bmatrix} \approx \begin{bmatrix} 1.00000 \\ 0.92308 \end{bmatrix} \\ A\mathbf{x}_2 &\approx \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1.00000 \\ 0.92308 \end{bmatrix} \approx \begin{bmatrix} 4.84615 \\ 4.76923 \end{bmatrix} & \mathbf{x}_3 &= \frac{A\mathbf{x}_2}{\max(A\mathbf{x}_2)} \approx \frac{1}{4.84615} \begin{bmatrix} 4.84615 \\ 4.76923 \end{bmatrix} \approx \begin{bmatrix} 1.00000 \\ 0.98413 \end{bmatrix} \\ A\mathbf{x}_3 &\approx \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1.00000 \\ 0.98413 \end{bmatrix} \approx \begin{bmatrix} 4.96825 \\ 4.95238 \end{bmatrix} & \mathbf{x}_4 &= \frac{A\mathbf{x}_3}{\max(A\mathbf{x}_3)} \approx \frac{1}{4.96825} \begin{bmatrix} 4.96825 \\ 4.95238 \end{bmatrix} \approx \begin{bmatrix} 1.00000 \\ 0.99681 \end{bmatrix} \\ A\mathbf{x}_4 &\approx \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1.00000 \\ 0.99681 \end{bmatrix} \approx \begin{bmatrix} 4.99361 \\ 4.99042 \end{bmatrix} & \mathbf{x}_5 &= \frac{A\mathbf{x}_4}{\max(A\mathbf{x}_4)} \approx \frac{1}{4.99361} \begin{bmatrix} 4.99361 \\ 4.99042 \end{bmatrix} \approx \begin{bmatrix} 1.00000 \\ 0.99936 \end{bmatrix} \end{aligned}$$

$$\lambda^{(1)} = \frac{A\mathbf{x}_1 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1} = \frac{(A\mathbf{x}_1)^T \mathbf{x}_1}{\mathbf{x}_1^T \mathbf{x}_1} \approx \frac{7.00000}{1.44444} \approx 4.84615$$

$$\lambda^{(2)} = \frac{A\mathbf{x}_2 \cdot \mathbf{x}_2}{\mathbf{x}_2 \cdot \mathbf{x}_2} = \frac{(A\mathbf{x}_2)^T \mathbf{x}_2}{\mathbf{x}_2^T \mathbf{x}_2} \approx \frac{9.24852}{1.85207} \approx 4.99361$$

$$\lambda^{(3)} = \frac{A\mathbf{x}_3 \cdot \mathbf{x}_3}{\mathbf{x}_3 \cdot \mathbf{x}_3} = \frac{(A\mathbf{x}_3)^T \mathbf{x}_3}{\mathbf{x}_3^T \mathbf{x}_3} \approx \frac{9.84203}{1.96851} \approx 4.99974$$

$$\lambda^{(4)} = \frac{A\mathbf{x}_4 \cdot \mathbf{x}_4}{\mathbf{x}_4 \cdot \mathbf{x}_4} = \frac{(A\mathbf{x}_4)^T \mathbf{x}_4}{\mathbf{x}_4^T \mathbf{x}_4} \approx \frac{9.96808}{1.99362} \approx 4.99999$$

$$\lambda^{(5)} = \frac{A\mathbf{x}_5 \cdot \mathbf{x}_5}{\mathbf{x}_5 \cdot \mathbf{x}_5} = \frac{(A\mathbf{x}_5)^T \mathbf{x}_5}{\mathbf{x}_5^T \mathbf{x}_5} \approx \frac{9.99360}{1.99872} \approx 5.00000$$

Thus, $\lambda^{(5)}$ approximates the dominant eigenvalue correctly to five decimal places and $\mathbf{x}_5$ closely approximates the dominant eigenvector

$$\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

that results by taking $t = 1$ in (6).

Whereas the power method with Euclidean scaling produces a sequence that approaches a *unit* dominant eigenvector, maximum entry scaling produces a sequence that approaches an eigenvector whose largest *component* is 1.

Rate of Convergence

If A is a symmetric matrix whose distinct eigenvalues can be arranged so that

$$|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \cdots \geq |\lambda_k|$$

then the “rate” at which the Rayleigh quotients converge to the dominant eigenvalue λ_1 depends on the ratio $|\lambda_1|/|\lambda_2|$; that is, the convergence is slow when this ratio is near 1 and rapid when it is large—the greater the ratio, the more rapid the convergence. For example, if A is a 2×2 symmetric matrix, then the greater the ratio $|\lambda_1|/|\lambda_2|$, the greater the disparity between the scaling effects of λ_1 and λ_2 in Figure 9.2.1, and hence the greater the effect that multiplication by A has on pulling the iterates toward the eigenspace of λ_1 . Indeed, the rapid convergence in Example 3 is due to the fact that $|\lambda_1|/|\lambda_2| = 5/1 = 5$, which is

considered to be a large ratio. In cases where the ratio is close to 1, the convergence of the power method may be so slow that other methods must be used.

Stopping Procedures

If λ is the exact value of the dominant eigenvalue, and if a power method produces the approximation $\lambda^{(k)}$ at the k th iteration, then we call

$$\left| \frac{\lambda - \lambda^{(k)}}{\lambda} \right| \quad (12)$$

the **relative error** in $\lambda^{(k)}$. Expressed as a percentage it is called the **percentage error** in $\lambda^{(k)}$. For example, if $\lambda = 5$ and the approximation after three iterations is $\lambda^{(3)} = 5.1$, then

$$\text{relative error in } \lambda^{(3)} = \left| \frac{\lambda - \lambda^{(3)}}{\lambda} \right| = \left| \frac{5 - 5.1}{5} \right| = |-0.02| = 0.02$$

$$\text{percentage error in } \lambda^{(3)} = 0.02 \times 100\% = 2\%$$

In applications one usually knows the relative error E that can be tolerated in the dominant eigenvalue, so the goal is to stop computing iterates once the relative error in the approximation to that eigenvalue is less than E . However, there is a problem in computing the relative error from (12) in that the eigenvalue λ is unknown. To circumvent this problem, it is usual to estimate λ by $\lambda^{(k)}$ and stop the computations when

$$\left| \frac{\lambda^{(k)} - \lambda^{(k-1)}}{\lambda^{(k)}} \right| < E \quad (13)$$

The quantity on the left side of (13) is called the **estimated relative error** in $\lambda^{(k)}$ and its percentage form is called the **estimated percentage error** in $\lambda^{(k)}$.

EXAMPLE 4 | Estimated Relative Error

For the computations in Example 3, find the smallest value of k for which the estimated percentage error in $\lambda^{(k)}$ is less than 0.1%.

Solution The estimated percentage errors in the approximations in Example 3 are as follows:

	Approximation	Relative Error	Percentage Error
$\lambda^{(2)}$:	$\left \frac{\lambda^{(2)} - \lambda^{(1)}}{\lambda^{(2)}} \right \approx \left \frac{4.99361 - 4.84615}{4.99361} \right $	≈ 0.02953	$= 2.953\%$
$\lambda^{(3)}$:	$\left \frac{\lambda^{(3)} - \lambda^{(2)}}{\lambda^{(3)}} \right \approx \left \frac{4.99974 - 4.99361}{4.99974} \right $	≈ 0.00123	$= 0.123\%$
$\lambda^{(4)}$:	$\left \frac{\lambda^{(4)} - \lambda^{(3)}}{\lambda^{(4)}} \right \approx \left \frac{4.99999 - 4.99974}{4.99999} \right $	≈ 0.00005	$= 0.005\%$
$\lambda^{(5)}$:	$\left \frac{\lambda^{(5)} - \lambda^{(4)}}{\lambda^{(5)}} \right \approx \left \frac{5.00000 - 4.99999}{5.00000} \right $	≈ 0.00000	$= 0\%$

Thus, $\lambda^{(4)} = 4.99999$ is the first approximation whose estimated percentage error is less than 0.1%.

Remark A rule for terminating an iterative process is called a **stopping procedure**. In the exercises, we will discuss stopping procedures for the power method that are based on the dominant eigenvector rather than the dominant eigenvalue.

Exercise Set 9.2

In Exercises 1–2, the distinct eigenvalues of a matrix are given. Determine whether A has a dominant eigenvalue, and if so, find it.

1. a. $\lambda_1 = 7, \lambda_2 = 3, \lambda_3 = -8, \lambda_4 = 1$
b. $\lambda_1 = -5, \lambda_2 = 3, \lambda_3 = 2, \lambda_4 = 5$
2. a. $\lambda_1 = 1, \lambda_2 = 0, \lambda_3 = -3, \lambda_4 = 2$
b. $\lambda_1 = -3, \lambda_2 = -2, \lambda_3 = -1, \lambda_4 = 3$

In Exercises 3–4, apply the power method with Euclidean scaling to the matrix A , starting with $\mathbf{x}_0$ and stopping at $\mathbf{x}_4$. Compare the resulting approximations to the exact values of the dominant eigenvalue and the corresponding unit eigenvector.

3. $A = \begin{bmatrix} 5 & -1 \\ -1 & -1 \end{bmatrix}; \mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
4. $A = \begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}; \mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

In Exercises 5–6, apply the power method with maximum entry scaling to the matrix A , starting with $\mathbf{x}_0$ and stopping at $\mathbf{x}_4$. Compare the resulting approximations to the exact values of the dominant eigenvalue and the corresponding scaled eigenvector.

5. $A = \begin{bmatrix} 1 & -3 \\ -3 & 5 \end{bmatrix}; \mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
6. $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 4 \end{bmatrix}; \mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

7. Let

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}; \mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

- a. Use the power method with maximum entry scaling to approximate a dominant eigenvector of A . Start with $\mathbf{x}_0$, round off all computations to three decimal places, and stop after three iterations.
 - b. Use the result in part (a) and the Rayleigh quotient to approximate the dominant eigenvalue of A .
 - c. Find the exact values of the eigenvector and eigenvalue approximated in parts (a) and (b).
 - d. Find the percentage error in the approximation of the dominant eigenvalue.
8. Repeat the directions of Exercise 7 with

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 10 \end{bmatrix}; \mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

In Exercises 9–10, a matrix A with a dominant eigenvalue and a sequence $\mathbf{x}_0, A\mathbf{x}_0, \dots, A^5\mathbf{x}_0$ are given. Use Formulas (9) and (10) to approximate the dominant eigenvalue and a corresponding eigenvector.

9. $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}; \mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, A\mathbf{x}_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, A^2\mathbf{x}_0 = \begin{bmatrix} 5 \\ 4 \end{bmatrix},$
 $A^3\mathbf{x}_0 = \begin{bmatrix} 13 \\ 14 \end{bmatrix}, A^4\mathbf{x}_0 = \begin{bmatrix} 41 \\ 40 \end{bmatrix}, A^5\mathbf{x}_0 = \begin{bmatrix} 121 \\ 122 \end{bmatrix}$
10. $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}; \mathbf{x}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, A\mathbf{x}_0 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, A^2\mathbf{x}_0 = \begin{bmatrix} 4 \\ 5 \end{bmatrix},$
 $A^3\mathbf{x}_0 = \begin{bmatrix} 14 \\ 13 \end{bmatrix}, A^4\mathbf{x}_0 = \begin{bmatrix} 40 \\ 41 \end{bmatrix}, A^5\mathbf{x}_0 = \begin{bmatrix} 122 \\ 121 \end{bmatrix}$

11. Consider matrices

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{x}_0 = \begin{bmatrix} a \\ b \end{bmatrix}$$

where $\mathbf{x}_0$ is a unit vector and $a \neq 0$. Show that even though the matrix A is symmetric and has a dominant eigenvalue, the power sequence (1) in Theorem 9.2.1 does not converge. This shows that the requirement in that theorem that the dominant eigenvalue be positive is essential.

12. Use the power method with Euclidean scaling to approximate the dominant eigenvalue and a corresponding eigenvector of A . Choose your own starting vector, and stop when the estimated percentage error in the eigenvalue approximation is less than 0.1%.

$$\text{a. } \begin{bmatrix} 1 & 3 & 3 \\ 3 & 4 & -1 \\ 3 & -1 & 10 \end{bmatrix} \quad \text{b. } \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 2 & -1 & 1 \\ 1 & -1 & 4 & 1 \\ 1 & 1 & 1 & 8 \end{bmatrix}$$

13. Repeat Exercise 12, but this time stop when all corresponding entries in two successive eigenvector approximations differ by less than 0.01 in absolute value.
14. Repeat Exercise 12 using maximum entry scaling.

Working with Proofs

15. Prove: If A is a nonzero $n \times n$ matrix, then $A^T A$ and $A A^T$ have positive dominant eigenvalues.
16. (For readers familiar with proof by induction) Let A be an $n \times n$ matrix, let $\mathbf{x}_0$ be a unit vector in R^n , and define the sequence $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k, \dots$ by

$$\mathbf{x}_1 = \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|}, \quad \mathbf{x}_2 = \frac{A\mathbf{x}_1}{\|A\mathbf{x}_1\|}, \quad \dots, \quad \mathbf{x}_k = \frac{A\mathbf{x}_{k-1}}{\|A\mathbf{x}_{k-1}\|}, \quad \dots$$

Prove by induction that $\mathbf{x}_k = A^k \mathbf{x}_0 / \|A^k \mathbf{x}_0\|$.

17. (For readers familiar with proof by induction) Let A be an $n \times n$ matrix, let $\mathbf{x}_0$ be a nonzero vector in R^n , and define the sequence $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k, \dots$ by

$$\mathbf{x}_1 = \frac{A\mathbf{x}_0}{\max(A\mathbf{x}_0)}, \quad \mathbf{x}_2 = \frac{A\mathbf{x}_1}{\max(A\mathbf{x}_1)}, \dots, \quad \mathbf{x}_k = \frac{A\mathbf{x}_{k-1}}{\max(A\mathbf{x}_{k-1})}, \dots$$

Prove by induction that

$$\mathbf{x}_k = \frac{A^k \mathbf{x}_0}{\max(A^k \mathbf{x}_0)}$$

Working with Technology

- T1. Use your technology utility to duplicate the computations in Example 2.
- T2. Use your technology utility to duplicate the computations in Example 3.

9.3 Comparison of Procedures for Solving Linear Systems

There is an old saying that “time is money.” This is especially true in industry where the cost of solving a linear system is generally determined by the time it takes for a computer to perform the required computations. This typically depends both on the speed of the computer processor and on the number of operations required by the algorithm. Thus, choosing the right algorithm has important financial implication in an industrial or research setting. In this section we will discuss some of the factors that affect the choice of algorithms for solving large-scale linear systems.

Flops and the Cost of Solving a Linear System

In computer jargon, an arithmetic operation (+, −, *, ÷) on two real numbers is called a **flop**, which is an acronym for “floating-point operation.”* The total number of flops required to solve a problem, which is called the **cost** of the solution, provides a convenient way of choosing between various algorithms for solving the problem. When needed, the cost in flops can be converted to units of time or money if the speed of the computer processor and the financial aspects of its operation are known. For example, today’s fastest computers are capable of performing in excess of 17 petaflops/s (1 petaflop = 10^{15} flops). Thus, an algorithm that costs 1,000,000 flops would be performed in 0.000000001 second. By contrast, today’s personal computers can perform in excess of 80 gigaflops/s (1 gigaflop = 10^9 flops). Thus, an algorithm that costs 1,000,000 flops would be performed on a personal computer in 0.0000125 second.

To illustrate how costs (in flops) can be computed, let us count the number of flops required to solve a linear system of n equations in n unknowns by Gauss–Jordan elimination. For this purpose we will need the following formulas for the sum of the first n positive integers and the sum of the squares of the first n positive integers:

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \quad (1)$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} \quad (2)$$

It is now common in computer jargon to write “FLOPs” to mean the number of “flops per second.” However, we will write “flops” simply as the plural of “flop.” When needed, we will write flops per second as flops/s.

*Real numbers are stored in computers as numerical approximations called **floating-point numbers**. In base 10, a floating-point number has the form $\pm d_1 d_2 \dots d_n \times 10^m$, where m is an integer, called the **mantissa**, and n is the number of digits to the right of the decimal point. The value of n varies with the computer. In some literature the term *flop* is used as a measure of processing speed and stands for “floating-point operations per second.” In our usage it is interpreted as a counting unit.

Let $Ax = b$ be a linear system of n equations in n unknowns to be solved by Gauss–Jordan elimination (or, equivalently, by Gaussian elimination with back substitution). For simplicity, let us assume that A is invertible and that no row interchanges are required to reduce the augmented matrix $[A \mid b]$ to row echelon form. The diagrams that accompany the following analysis provide a convenient way of counting the operations required to introduce a leading 1 in the first row and then zeros below it. In our operation counts, we will lump divisions and multiplications together as “multiplications,” and we will lump additions and subtractions together as “additions.”

Step 1. It requires n flops (multiplications) to introduce the leading 1 in the first row.

$$\left[\begin{array}{cccccc|c} 1 & \times & \times & \cdots & \times & \times & \times \\ \cdot & \cdot & \cdot & \cdots & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdots & \cdot & \cdot & \cdot \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ \cdot & \cdot & \cdot & \cdots & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdots & \cdot & \cdot & \cdot \end{array} \right] \quad \left[\begin{array}{l} \times \text{ denotes a quantity that is being computed.} \\ \cdot \text{ denotes a quantity that is not being computed.} \\ \text{The augmented matrix size is } n \times (n + 1). \end{array} \right]$$

Step 2. It requires n multiplications and n additions to introduce a zero below the leading 1, and there are $n - 1$ rows below the leading 1, so the number of flops required to introduce zeros below the leading 1 is $2n(n - 1)$.

$$\left[\begin{array}{cccccc|c} 1 & \cdot & \cdot & \cdots & \cdot & \cdot & \cdot \\ 0 & \times & \times & \cdots & \times & \times & \times \\ 0 & \times & \times & \cdots & \times & \times & \times \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & \times & \times & \cdots & \times & \times & \times \\ 0 & \times & \times & \cdots & \times & \times & \times \end{array} \right]$$

Column 1. Combining Steps 1 and 2, the number of flops required for column 1 is

$$n + 2n(n - 1) = 2n^2 - n$$

Column 2. The procedure for column 2 is the same as for column 1, except that now we are dealing with one less row and one less column. Thus, the number of flops required to introduce the leading 1 in row 2 and the zeros below it can be obtained by replacing n by $n - 1$ in the flop count for the first column. Thus, the number of flops required for column 2 is

$$2(n - 1)^2 - (n - 1)$$

Column 3. By the argument for column 2, the number of flops required for column 3 is

$$2(n - 2)^2 - (n - 2)$$

Total. The pattern should now be clear. The total number of flops required to create the n leading 1's and the associated zeros is

$$(2n^2 - n) + [2(n - 1)^2 - (n - 1)] + [2(n - 2)^2 - (n - 2)] + \cdots + (2 - 1)$$

which we can rewrite as

$$2[n^2 + (n - 1)^2 + \cdots + 1] - [n + (n - 1) + \cdots + 1]$$

or on applying Formulas (1) and (2) as

$$2 \left[\frac{n(n + 1)(2n + 1)}{6} - \frac{n(n + 1)}{2} \right] = \frac{2}{3}n^3 + \frac{1}{2}n^2 - \frac{1}{6}n$$

Next, let us count the number of operations required to complete the backward phase (the back substitution).

Column n . It requires $n - 1$ multiplications and $n - 1$ additions to introduce zeros above the leading 1 in the n th column, so the total number of flops required for the column is $2(n - 1)$.

$$\left[\begin{array}{cccccc|c} 1 & \bullet & \bullet & \cdots & \bullet & 0 & \times \\ 0 & 1 & \bullet & \cdots & \bullet & 0 & \times \\ 0 & 0 & 1 & \cdots & \bullet & 0 & \times \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & \times \\ 0 & 0 & 0 & \cdots & 0 & 1 & \bullet \end{array} \right]$$

Column $(n - 1)$. The procedure is the same as for Step 1, except that now we are dealing with one less row. Thus, the number of flops required for the $(n - 1)$ st column is $2(n - 2)$.

$$\left[\begin{array}{cccccc|c} 1 & \bullet & \bullet & \cdots & 0 & 0 & \times \\ 0 & 1 & \bullet & \cdots & 0 & 0 & \times \\ 0 & 0 & 1 & \cdots & 0 & 0 & \times \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 & \bullet \\ 0 & 0 & 0 & \cdots & 0 & 1 & \bullet \end{array} \right]$$

Column $(n - 2)$. By the argument for column $(n - 1)$, the number of flops required for column $(n - 2)$ is $2(n - 3)$.

Total. The pattern should now be clear. The total number of flops to complete the backward phase is

$$2(n - 1) + 2(n - 2) + 2(n - 3) + \cdots + 2(n - n) = 2[n^2 - (1 + 2 + \cdots + n)]$$

which we can rewrite using Formula (1) as

$$2\left(n^2 - \frac{n(n + 1)}{2}\right) = n^2 - n$$

In summary, we have shown that for Gauss–Jordan elimination the number of flops required for the forward and backward phases is

$$\text{flops for forward phase} = \frac{2}{3}n^3 + \frac{1}{2}n^2 - \frac{1}{6}n \quad (3)$$

$$\text{flops for backward phase} = n^2 - n \quad (4)$$

Thus, the total cost of solving a linear system by Gauss–Jordan elimination is

$$\text{flops for both phases} = \frac{2}{3}n^3 + \frac{3}{2}n^2 - \frac{7}{6}n \quad (5)$$

Cost Estimates for Solving Large Linear Systems

It is a property of polynomials that for large values of the independent variable the term of highest power makes the major contribution to the value of the polynomial. Thus, for *large* linear systems we can use (3) and (4) to approximate the number of flops in the forward and backward phases as

$$\text{flops for forward phase} \approx \frac{2}{3}n^3 \quad (6)$$

$$\text{flops for backward phase} \approx n^2 \quad (7)$$

This shows that it is more costly to execute the forward phase than the backward phase for large linear systems. Indeed, the cost difference between the forward and backward phases can be enormous, as the next example shows.

We leave it as an exercise for you to confirm the results in [Table 1](#).

TABLE 1

Approximate Cost for an $n \times n$ Matrix A with Large n	
Algorithm	Cost in Flops
Gauss–Jordan elimination (forward phase)	$\approx \frac{2}{3}n^3$
Gauss–Jordan elimination (backward phase)	$\approx n^2$
LU -decomposition of A	$\approx \frac{2}{3}n^3$
Forward substitution to solve $Ly = \mathbf{b}$	$\approx n^2$
Backward substitution to solve $U\mathbf{x} = \mathbf{y}$	$\approx n^2$
A^{-1} by reducing $[A \mid I]$ to $[I \mid A^{-1}]$	$\approx 2n^3$
Compute $A^{-1}\mathbf{b}$	$\approx 2n^3$

The cost in flops for Gaussian elimination is the same as that for the forward phase of Gauss–Jordan elimination.

EXAMPLE 1 | Cost of Solving a Large Linear System

Approximate the time required to execute the forward and backward phases of Gauss–Jordan elimination for a system of one million ($= 10^6$) equations in one million unknowns using a computer that can execute 10 petaflops per second (1 petaflop $= 10^{15}$ flops).

Solution We have $n = 10^6$ for the given system, so from (6) and (7) the number of petaflops required for the forward and backward phases is

$$\text{petaflops for forward phase} \approx \frac{2}{3}n^3 \times 10^{-15} = \frac{2}{3}(10^6)^3 \times 10^{-15} = \frac{2}{3} \times 10^3$$

$$\text{petaflops for backward phase} \approx n^2 \times 10^{-15} = (10^6)^2 \times 10^{-15} = 10^{-3}$$

Thus, at 10 petaflops/s the execution times for the forward and backward phases are

$$\text{time for forward phase} \approx \left(\frac{2}{3} \times 10^3\right) \times 10^{-1} \text{ s} \approx 66.67 \text{ s}$$

$$\text{time for backward phase} \approx (10^{-3}) \times 10^{-1} \text{ s} \approx 0.0001 \text{ s}$$

Considerations in Choosing an Algorithm for Solving a Linear System

For a *single* linear system $A\mathbf{x} = \mathbf{b}$ of n equations in n unknowns, the methods of LU -decomposition and Gauss–Jordan elimination differ in bookkeeping but otherwise involve the same number of flops. Thus, neither method has a cost advantage over the other. However, LU -decomposition has the following advantages that make it the method of choice:

- Gauss–Jordan elimination and Gaussian elimination both use the augmented matrix $[A \mid \mathbf{b}]$, so $\mathbf{b}$ must be known. In contrast, LU -decomposition uses only the matrix A , so once that decomposition is known it can be used with as many right-hand sides as are required.
- The LU -decomposition that is computed to solve $A\mathbf{x} = \mathbf{b}$ can be used to compute A^{-1} , if needed, with little additional work.
- For large linear systems in which computer memory is at a premium, one can dispense with the storage of the 1's and zeros that appear on or below the main diagonal of U , since those entries are known from the form of U . The space that this opens up

can then be used to store the entries of L , thereby reducing the amount of memory required to solve the system.

- If A is a large matrix consisting mostly of zeros, and if the nonzero entries are concentrated in a “band” around the main diagonal, then there are techniques that can be used to reduce the cost of LU -decomposition, giving it an advantage over Gauss–Jordan elimination.

Exercise Set 9.3

1. A certain computer can execute 10 gigaflops per second. Use Formula (5) to find the time required to solve the system using Gauss–Jordan elimination.
 - a. A system of 1000 equations in 1000 unknowns.
 - b. A system of 10,000 equations in 10,000 unknowns.
 - c. A system of 100,000 equations in 100,000 unknowns.
2. A certain computer can execute 100 gigaflops per second. Use Formula (5) to find the time required to solve the system using Gauss–Jordan elimination.
 - a. A system of 10,000 equations in 10,000 unknowns.
 - b. A system of 100,000 equations in 100,000 unknowns.
 - c. A system of 1,000,000 equations in 1,000,000 unknowns.
3. A certain computer can execute 70 gigaflops per second. Use **Table 1** to estimate the time required to perform the following operations on the invertible $10,000 \times 10,000$ matrix A .
 - a. Execute the forward phase of Gauss–Jordan elimination.
 - b. Execute the backward phase of Gauss–Jordan elimination.
 - c. LU -decomposition of A .
 - d. Find A^{-1} by reducing $[A \mid I]$ to $[I \mid A^{-1}]$.
4. The IBM Sequoia computer can operate at speeds in excess of 16 petaflops per second (1 petaflop = 10^{15} flops). Use Table 1 to estimate the time required to perform the following operations on an invertible $100,000 \times 100,000$ matrix A .
 - a. Execute the forward phase of Gauss–Jordan elimination.
 - b. Execute the backward phase of Gauss–Jordan elimination.
 - c. LU -decomposition of A .
 - d. Find A^{-1} by reducing $[A \mid I]$ to $[I \mid A^{-1}]$.
5. a. Approximate the time required to execute the forward phase of Gauss–Jordan elimination for a system of 100,000 equations in 100,000 unknowns using a computer that can execute 1 gigaflop per second. Do the same for the backward phase. (See Table 1.)
 - b. How many gigaflops per second must a computer be able to execute to find the LU -decomposition of a matrix of size $10,000 \times 10,000$ in less than 0.5 s? (See Table 1.)
6. About how many teraflops per second must a computer be able to execute to find the inverse $100,000 \times 100,000$ matrix in less than 0.5 s? (1 teraflop = 10^{12} flops.)

In Exercises 7–10, suppose A and B are $n \times n$ matrices and c is a real number.

7. How many flops are required to compute cA ?
8. How many flops are required to compute $A + B$?
9. How many flops are required to compute AB ?
10. If A is a diagonal matrix and k is a positive integer, how many flops are required to compute A^k ?

9.4

Singular Value Decomposition

In this section we will discuss an extension of the diagonalization theory for $n \times n$ symmetric matrices to general $m \times n$ matrices. The results that we will develop in this section have applications to compression, storage, and transmission of digitized information and form the basis for many of the best computational algorithms that are currently available for solving linear systems.

Decompositions of Square Matrices

We saw in Formula (2) of Section 7.2 that every symmetric matrix A with real entries can be expressed as

$$A = PDP^T \quad (1)$$

where P is an orthogonal matrix whose columns are eigenvectors of A , and D is the diagonal matrix whose diagonal entries are the eigenvalues corresponding to the column

vectors of P . In this section we will call (1) an **eigenvalue decomposition** of A (abbreviated EVD of A).

If an $n \times n$ matrix A is not symmetric, then it does not have an eigenvalue decomposition, but it does have a **Hessenberg decomposition**

$$A = PHP^T$$

in which P is an orthogonal matrix and H is in upper Hessenberg form (Theorem 7.2.4). Moreover, if A has real eigenvalues, then it has a **Schur decomposition**

$$A = PSP^T$$

in which P is an orthogonal matrix and S is upper triangular (Theorem 7.2.3).

The eigenvalue, Hessenberg, and Schur decompositions are important in numerical algorithms not only because the matrices D , H , and S have simpler forms than A , but also because the orthogonal matrices that appear in these factorizations do not magnify roundoff error. To see why this is so, suppose that $\hat{\mathbf{x}}$ is a column vector whose entries are known exactly and that

$$\mathbf{x} = \hat{\mathbf{x}} + \mathbf{e}$$

is the vector that results when roundoff error is present in the entries of $\hat{\mathbf{x}}$. If P is an orthogonal matrix, then the length-preserving property of orthogonal transformations implies that

$$\|P\mathbf{x} - P\hat{\mathbf{x}}\| = \|\mathbf{x} - \hat{\mathbf{x}}\| = \|\mathbf{e}\|$$

which tells us that the error in approximating $P\hat{\mathbf{x}}$ by $P\mathbf{x}$ has the same magnitude as the error in approximating $\hat{\mathbf{x}}$ by $\mathbf{x}$.

There are two main paths that one might follow in looking for other kinds of decompositions of a general square matrix A : One might look for decompositions of the form

$$A = PJP^{-1}$$

in which P is invertible but not necessarily orthogonal, or one might look for decompositions of the form

$$A = U\Sigma V^T$$

in which U and V are orthogonal but not necessarily the same. The first path leads to decompositions in which J is either diagonal or a certain kind of block diagonal matrix, called a **Jordan canonical form** in honor of the French mathematician Camille Jordan (see p. 538). Jordan canonical forms, which we will not consider in this text, are important theoretically and in certain applications, but they are of lesser importance numerically because of the roundoff difficulties that result from the lack of orthogonality in P . In this section we will focus on the second path.

Singular Values

Since matrix products of the form $A^T A$ will play an important role in our work, we will begin with two basic theorems about them.

Theorem 9.4.1

If A is an $m \times n$ matrix, then:

- (a) A and $A^T A$ have the same null space.
- (b) A and $A^T A$ have the same row space.
- (c) A^T and $A^T A$ have the same column space.
- (d) A and $A^T A$ have the same rank.

We will prove part (a) and leave the remaining proofs for the exercises.

Proof (a) We must show that every solution of $A\mathbf{x} = \mathbf{0}$ is a solution of $A^T A\mathbf{x} = \mathbf{0}$, and conversely. If $\mathbf{x}_0$ is any solution of $A\mathbf{x} = \mathbf{0}$, then $\mathbf{x}_0$ is also a solution of $A^T A\mathbf{x} = \mathbf{0}$ since

$$A^T A\mathbf{x}_0 = A^T (A\mathbf{x}_0) = A^T \mathbf{0} = \mathbf{0}$$

Conversely, if $\mathbf{x}_0$ is any solution of $A^T A\mathbf{x} = \mathbf{0}$, then $\mathbf{x}_0$ is in the null space of $A^T A$ and hence is orthogonal to all vectors in the row space of $A^T A$ by part (s) of Theorem 8.2.4. However, $A^T A$ is symmetric, so $\mathbf{x}_0$ is also orthogonal to every vector in the column space of $A^T A$. In particular, $\mathbf{x}_0$ must be orthogonal to the vector $(A^T A)\mathbf{x}_0$; that is,

$$\mathbf{x}_0 \cdot (A^T A)\mathbf{x}_0 = 0$$

Using the first row in Table 1 of Section 3.2 and properties of the transpose operation we can rewrite this as

$$\mathbf{x}_0^T (A^T A)\mathbf{x}_0 = (A\mathbf{x}_0)^T (A\mathbf{x}_0) = (A\mathbf{x}_0) \cdot (A\mathbf{x}_0) = \|A\mathbf{x}_0\|^2 = 0$$

which implies that $A\mathbf{x}_0 = \mathbf{0}$, thereby proving that $\mathbf{x}_0$ is a solution of $A\mathbf{x} = \mathbf{0}$. ■

Theorem 9.4.2

If A is an $m \times n$ matrix, then:

- (a) $A^T A$ is orthogonally diagonalizable.
- (b) The eigenvalues of $A^T A$ are nonnegative real numbers.

Proof (a) The matrix $A^T A$, being symmetric, is orthogonally diagonalizable by Theorem 7.2.1.

Proof (b) Since $A^T A$ is orthogonally diagonalizable, there is an orthonormal basis for R^n consisting of eigenvectors of $A^T A$, say $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. If we let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the corresponding eigenvalues, then for $1 \leq i \leq n$ we have

$$\begin{aligned} \|A\mathbf{v}_i\|^2 &= A\mathbf{v}_i \cdot A\mathbf{v}_i = \mathbf{v}_i \cdot A^T A\mathbf{v}_i && \text{[Formula (26) of Section 3.2]} \\ &= \mathbf{v}_i \cdot \lambda_i \mathbf{v}_i = \lambda_i (\mathbf{v}_i \cdot \mathbf{v}_i) = \lambda_i \|\mathbf{v}_i\|^2 = \lambda_i \end{aligned}$$

It follows from this relationship that $\lambda_i \geq 0$. ■

We will assume throughout this section that the eigenvalues of $A^T A$ are named so that

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$$

and hence that

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n \geq 0$$

Definition 1

If A is an $m \times n$ matrix, and if $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of $A^T A$, then the numbers

$$\sigma_1 = \sqrt{\lambda_1}, \quad \sigma_2 = \sqrt{\lambda_2}, \dots, \quad \sigma_n = \sqrt{\lambda_n}$$

are called the **singular values** of A .

EXAMPLE 1 | Singular Values

Find the singular values of the matrix

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Solution The first step is to find the eigenvalues of the matrix

$$A^T A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

The characteristic polynomial of $A^T A$ is

$$\lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$$

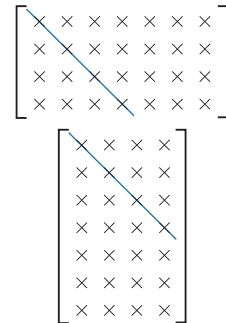
so the eigenvalues of $A^T A$ are $\lambda_1 = 3$ and $\lambda_2 = 1$ and the singular values of A in order of decreasing size are

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{3}, \quad \sigma_2 = \sqrt{\lambda_2} = 1$$

Singular Value Decomposition

Before turning to the main result in this section, we will find it useful to extend the notion of a “main diagonal” to matrices that are not square. We define the **main diagonal** of an $m \times n$ matrix to be the line of entries shown in **Figure 9.4.1**—it starts at the upper left corner and extends diagonally as far as it can go. We will refer to the entries on the main diagonal as the **diagonal entries**.

We are now ready to consider the main result in this section, which is concerned with a specific way of factoring a general $m \times n$ matrix A . This factorization, called **singular value decomposition** (abbreviated SVD) will be given in two forms, a brief form that captures the main idea, and an expanded form that spells out the details. The proof is given at the end of this section.



Main diagonals

FIGURE 9.4.1

Theorem 9.4.3

Singular Value Decomposition (Brief Form)

If A is an $m \times n$ matrix of rank k , then A can be expressed in the form $A = U\Sigma V^T$, where Σ has size $m \times n$ and can be expressed in partitioned form as

$$\Sigma = \left[\begin{array}{c|c} D & 0_{k \times (n-k)} \\ \hline 0_{(m-k) \times k} & 0_{(m-k) \times (n-k)} \end{array} \right]$$

in which D is a diagonal $k \times k$ matrix whose successive entries are the first k singular values of A in nonincreasing order, U is an $m \times m$ orthogonal matrix, and V is an $n \times n$ orthogonal matrix.

Historical Note



Harry Bateman
(1882–1946)

The term *singular value* is apparently due to the British-born mathematician Harry Bateman, who used it in a research paper published in 1908. Bateman emigrated to the United States in 1910, teaching at Bryn Mawr College, Johns Hopkins University, and finally at the California Institute of Technology. Interestingly, he was awarded his Ph.D. in 1913 by Johns Hopkins at which point in time he was already an eminent mathematician with 60 publications to his name.

[Image: Courtesy of the Archives, California Institute of Technology]

Theorem 9.4.4

Singular Value Decomposition (Expanded Form)

If A is an $m \times n$ matrix of rank k , then A can be factored as

$$A = U\Sigma V^T = [\mathbf{u}_1 \ \mathbf{u}_2 \ \cdots \ \mathbf{u}_k \mid \mathbf{u}_{k+1} \ \cdots \ \mathbf{u}_m] \left[\begin{array}{cccc|ccc} \sigma_1 & 0 & \cdots & 0 & & & \\ 0 & \sigma_2 & \cdots & 0 & & & \\ \vdots & \vdots & \ddots & \vdots & & & \\ 0 & 0 & \cdots & \sigma_k & & & \\ \hline & & & & 0_{(m-k) \times k} & & \\ & & & & & 0_{(m-k) \times (n-k)} & \end{array} \right] \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_k^T \\ \hline \mathbf{v}_{k+1}^T \\ \vdots \\ \mathbf{v}_n^T \end{bmatrix}$$

in which U , Σ , and V have sizes $m \times m$, $m \times n$, and $n \times n$, respectively, and in which:

- (a) $V = [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n]$ orthogonally diagonalizes $A^T A$.
- (b) The nonzero diagonal entries of Σ are $\sigma_1 = \sqrt{\lambda_1}$, $\sigma_2 = \sqrt{\lambda_2}$, $\dots$, $\sigma_k = \sqrt{\lambda_k}$, where $\lambda_1, \lambda_2, \dots, \lambda_k$ are the nonzero eigenvalues of $A^T A$ corresponding to the column vectors of V .
- (c) The column vectors of V are ordered so that $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k > 0$.
- (d) $\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|} = \frac{1}{\sigma_i} A\mathbf{v}_i \quad (i = 1, 2, \dots, k)$
- (e) $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ is an orthonormal basis for $\text{col}(A)$.
- (f) $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k, \mathbf{u}_{k+1}, \dots, \mathbf{u}_m\}$ is an extension of $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ to an orthonormal basis for R^m .

The vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are called the **left singular vectors** of A , and the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are called the **right singular vectors** of A .

EXAMPLE 2 | Singular Value Decomposition if A Is Not Square

Find a singular value decomposition of the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Solution We showed in Example 1 that the eigenvalues of $A^T A$ are $\lambda_1 = 3$ and $\lambda_2 = 1$ and that the corresponding singular values of A are $\sigma_1 = \sqrt{3}$ and $\sigma_2 = 1$. We leave it for you to verify that

$$\mathbf{v}_1 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix}$$

are eigenvectors corresponding to λ_1 and λ_2 , respectively, and that $V = [\mathbf{v}_1 \mid \mathbf{v}_2]$ orthogonally diagonalizes $A^T A$. From part (d) of Theorem 9.4.4, the vectors

$$\mathbf{u}_1 = \frac{1}{\sigma_1} A\mathbf{v}_1 = \frac{\sqrt{3}}{3} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{6}}{3} \\ \frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\mathbf{u}_2 = \frac{1}{\sigma_2} A\mathbf{v}_2 = (1) \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

are two of the three column vectors of U . Note that $\mathbf{u}_1$ and $\mathbf{u}_2$ are orthonormal, as expected. We could extend the set $\{\mathbf{u}_1, \mathbf{u}_2\}$ to an orthonormal basis for $\mathbb{R}^3$. However, the computations will be easier if we first remove the messy radicals by multiplying $\mathbf{u}_1$ and $\mathbf{u}_2$ by appropriate scalars. Thus, we will look for a unit vector $\mathbf{u}_3$ that is orthogonal to

$$\sqrt{6}\mathbf{u}_1 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \sqrt{2}\mathbf{u}_2 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

To satisfy these two orthogonality conditions, the vector $\mathbf{u}_3$ must be a solution of the homogeneous linear system

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

We leave it for you to show that a general solution of this system is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

Normalizing the vector on the right yields

$$\mathbf{u}_3 = \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$$

Thus, the singular value decomposition of A is

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{6}}{3} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} & \frac{1}{\sqrt{3}} \\ \frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}$$

$A \quad = \quad U \quad \Sigma \quad V^T$

You may want to confirm the validity of this equation by multiplying out the matrices on the right side.

OPTIONAL: We conclude this section with an optional proof of Theorem 9.4.4.

Proof of Theorem 9.4.4 For notational simplicity we will prove this theorem in the case where A is an $n \times n$ matrix. To modify the argument for an $m \times n$ matrix you need only make the notational adjustments required to account for the possibility that $m > n$ or $n > m$.

The matrix $A^T A$ is symmetric, so it has an eigenvalue decomposition

$$A^T A = V D V^T$$

in which the column vectors of

$$V = [\mathbf{v}_1 \mid \mathbf{v}_2 \mid \cdots \mid \mathbf{v}_n]$$

are unit eigenvectors of $A^T A$, and D is a diagonal matrix whose successive diagonal entries $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of $A^T A$ corresponding in succession to the column vectors of V . Since A is assumed to have rank k , it follows from Theorem 9.4.1 that $A^T A$ also

has rank k . It follows as well that D has rank k , since it is similar to $A^T A$ and rank is a similarity invariant. Thus, the diagonal matrix D can be expressed in the form

$$D = \begin{bmatrix} \lambda_1 & & & & 0 \\ & \lambda_2 & & & \\ & & \ddots & & \\ & & & \lambda_k & \\ 0 & & & & 0 & \ddots & \\ & & & & & & 0 \end{bmatrix} \quad (2)$$

where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k > 0$. Now let us consider the set of image vectors

$$\{A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_n\} \quad (3)$$

This is an orthogonal set, for if $i \neq j$, then the orthogonality of $\mathbf{v}_i$ and $\mathbf{v}_j$ implies that

$$A\mathbf{v}_i \cdot A\mathbf{v}_j = \mathbf{v}_i \cdot A^T A \mathbf{v}_j = \mathbf{v}_i \cdot \lambda_j \mathbf{v}_j = \lambda_j (\mathbf{v}_i \cdot \mathbf{v}_j) = 0$$

Moreover, the first k vectors in (3) are nonzero since we showed in the proof of Theorem 9.4.2(b) that $\|A\mathbf{v}_i\|^2 = \lambda_i$ for $i = 1, 2, \dots, n$, and we have assumed that the first k diagonal entries in (2) are positive. Thus,

$$S = \{A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_k\}$$

is an orthogonal set of *nonzero* vectors in the column space of A . But the column space of A has dimension k since

$$\text{rank}(A) = \text{rank}(A^T A) = k$$

and hence S , being a linearly independent set of k vectors, must be an orthogonal basis for $\text{col}(A)$. If we now normalize the vectors in S , we will obtain an orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ for $\text{col}(A)$ in which

$$\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|} = \frac{1}{\sqrt{\lambda_i}} A\mathbf{v}_i \quad (1 \leq i \leq k)$$

or, equivalently, in which

$$A\mathbf{v}_1 = \sqrt{\lambda_1} \mathbf{u}_1 = \sigma_1 \mathbf{u}_1, \quad A\mathbf{v}_2 = \sqrt{\lambda_2} \mathbf{u}_2 = \sigma_2 \mathbf{u}_2, \dots, \quad A\mathbf{v}_k = \sqrt{\lambda_k} \mathbf{u}_k = \sigma_k \mathbf{u}_k \quad (4)$$

It follows from Theorem 6.3.6 that we can extend this to an orthonormal basis

$$\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k, \mathbf{u}_{k+1}, \dots, \mathbf{u}_n\}$$

for R^n . Now let U be the orthogonal matrix

$$U = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_k \quad \mathbf{u}_{k+1} \quad \cdots \quad \mathbf{u}_n]$$

and let Σ be the diagonal matrix

$$\Sigma = \begin{bmatrix} \sigma_1 & & & & 0 \\ & \sigma_2 & & & \\ & & \ddots & & \\ & & & \sigma_k & \\ & & & & 0 & \ddots \\ 0 & & & & & & 0 \end{bmatrix}$$

It follows from (4), and the fact that $A\mathbf{v}_i = 0$ for $i > k$, that

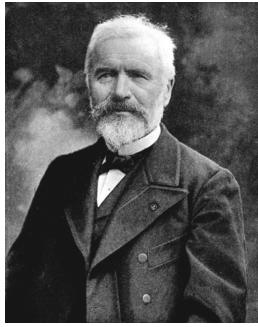
$$\begin{aligned} U\Sigma &= [\sigma_1 \mathbf{u}_1 \quad \sigma_2 \mathbf{u}_2 \quad \cdots \quad \sigma_k \mathbf{u}_k \quad 0 \quad \cdots \quad 0] \\ &= [A\mathbf{v}_1 \quad A\mathbf{v}_2 \quad \cdots \quad A\mathbf{v}_k \quad A\mathbf{v}_{k+1} \quad \cdots \quad A\mathbf{v}_n] \\ &= AV \end{aligned}$$

which we can rewrite using the orthogonality of V as $A = U\Sigma V^T$. ■

Historical Note



Eugenio Beltrami
(1835–1900)



Camille Jordan
(1838–1922)



Herman Klaus Weyl
(1885–1955)



Gene H. Golub
(1932–2007)

The theory of singular value decompositions can be traced back to the work of five people: the Italian mathematician Eugenio Beltrami, the French mathematician Camille Jordan, the English mathematician James Sylvester (see p. 36), and the German mathematicians Erhard Schmidt (see p. 369) and the mathematician Herman Weyl. More recently, the pioneering efforts of the American mathematician Gene Golub produced a stable and efficient algorithm for computing it. Beltrami and Jordan were the progenitors of the decomposition—Beltrami gave a proof of the result for real, invertible matrices with distinct singular values in 1873. Subsequently, Jordan refined the theory and eliminated the unnecessary restrictions imposed by Beltrami. Sylvester, apparently unfamiliar with the work of Beltrami and Jordan, rediscovered the result in 1889 and suggested its importance. Schmidt was the first person to show that the singular value decomposition could be used to approximate a matrix by another matrix with lower rank, and, in so doing, he transformed it from a mathematical curiosity to an important practical tool. Weyl showed how to find the lower rank approximations in the presence of error.

[Images: <http://www-history.mcs.st-andrews.ac.uk/history/PictDisplay/Beltrami.html> (Beltrami); The Granger Collection, New York (Jordan); Courtesy Electronic Publishing Services, Inc., New York City (Weyl); Courtesy of Hector Garcia-Molina (Golub)]

Exercise Set 9.4

In Exercises 1–4, find the distinct singular values of A .

1. $A = \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$

2. $A = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$

3. $A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$

4. $A = \begin{bmatrix} \sqrt{2} & 0 \\ 1 & \sqrt{2} \end{bmatrix}$

In Exercises 5–12, find a singular value decomposition of A .

5. $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

6. $A = \begin{bmatrix} -3 & 0 \\ 0 & -4 \end{bmatrix}$

7. $A = \begin{bmatrix} 4 & 6 \\ 0 & 4 \end{bmatrix}$

8. $A = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$

$$9. A = \begin{bmatrix} -2 & 2 \\ -1 & 1 \\ 2 & -2 \end{bmatrix}$$

$$10. A = \begin{bmatrix} -2 & -1 & 2 \\ 2 & 1 & -2 \end{bmatrix}$$

$$11. A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$12. A = \begin{bmatrix} 6 & 4 \\ 0 & 0 \\ 4 & 0 \end{bmatrix}$$

Working with Proofs

13. Prove: If A is an $m \times n$ matrix, then $A^T A$ and $A A^T$ have the same rank.
14. Prove part (d) of Theorem 9.4.1 by using part (a) of the theorem and the fact that A and $A^T A$ have n columns.
15. a. Prove part (b) of Theorem 9.4.1 by first showing that $\text{row}(A^T A)$ is a subspace of $\text{row}(A)$.
b. Prove part (c) of Theorem 9.4.1 by using part (b).
16. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation whose standard matrix A has the singular value decomposition $A = U \Sigma V^T$, and let $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and $B' = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ be the column vectors of V and U , respectively. Prove that

$$\Sigma = [T]_{B', B}$$

17. Prove that the singular values of $A^T A$ are the squares of the singular values of A .
18. Prove that if $A = U \Sigma V^T$ is a singular value decomposition of A , then U orthogonally diagonalizes $A A^T$.
19. A **polar decomposition** of an $n \times n$ matrix A is a factorization $A = PQ$ in which P is a positive semidefinite $n \times n$ matrix with the same rank as A , and Q is an orthogonal $n \times n$ matrix.

- a. Prove that if $A = U \Sigma V^T$ is the singular value decomposition of A , then $A = (U \Sigma U^T)(U V^T)$ is a polar decomposition of A .
- b. Find a polar decomposition of the matrix in Exercise 5.

True-False Exercises

- TF. In parts (a)–(g) determine whether the statement is true or false, and justify your answer.
- a. If A is an $m \times n$ matrix, then $A^T A$ is an $m \times m$ matrix.
- b. If A is an $m \times n$ matrix, then $A^T A$ is a symmetric matrix.
- c. If A is an $m \times n$ matrix, then the eigenvalues of $A^T A$ are positive real numbers.
- d. If A is an $n \times n$ matrix, then A is orthogonally diagonalizable.
- e. If A is an $m \times n$ matrix, then $A^T A$ is orthogonally diagonalizable.
- f. The eigenvalues of $A^T A$ are the singular values of A .
- g. Every $m \times n$ matrix has a singular value decomposition.

Working with Technology

- T1. Use your technology utility to duplicate the computations in Example 2.
- T2. For the given matrix A , use the steps in Example 2 to find matrices U , Σ , and V^T in a singular value decomposition $A = U \Sigma V^T$.

$$a. A = \begin{bmatrix} -2 & -1 & 2 \\ 2 & 1 & -2 \end{bmatrix} \quad b. A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ -1 & 1 \end{bmatrix}$$

9.5

Data Compression Using Singular Value Decomposition

Efficient transmission and storage of large quantities of digital data has become a major problem in our technological world. In this section we will discuss the role that singular value decomposition plays in compressing digital data so that it can be transmitted more rapidly and stored in less space. We assume here that you have read Section 9.4.

Reduced Singular Value Decomposition

Algebraically, the zero rows and columns of the matrix Σ in Theorem 9.4.4 are superfluous and can be eliminated by multiplying out the expression $U \Sigma V^T$ using block multiplication

and the partitioning shown in that formula. The products that involve zero blocks as factors drop out, leaving

$$A = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_k] \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_k \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_k^T \end{bmatrix} \quad (1)$$

which is called a **reduced singular value decomposition** of A . In this text we will denote the matrices on the right side of (1) by U_1, Σ_1 , and V_1^T , respectively, and we will write this equation as

$$A = U_1 \Sigma_1 V_1^T \quad (2)$$

Note that the sizes of U_1, Σ_1 , and V_1^T are $m \times k, k \times k$, and $k \times n$, respectively, and that the matrix Σ_1 is invertible since its diagonal entries are positive.

If we multiply out on the right side of (1) using the column-row rule, then we obtain

$$A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \cdots + \sigma_k \mathbf{u}_k \mathbf{v}_k^T \quad (3)$$

which is called a **reduced singular value expansion** of A . This result applies to *all* matrices, whereas the spectral decomposition [Formula (7) of Section 7.2] applies only to symmetric matrices.

Remark It can be proved that an $m \times n$ matrix M has rank 1 if and only if it can be factored as $M = \mathbf{u} \mathbf{v}^T$, where $\mathbf{u}$ is a column vector in R^m and $\mathbf{v}$ is a column vector in R^n . Thus, a reduced singular value decomposition expresses a matrix A of rank k as a linear combination of k rank 1 matrices.

EXAMPLE 1 | Reduced Singular Value Decomposition

Find a reduced singular value decomposition and a reduced singular value expansion of the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Solution In Example 2 of Section 9.4 we found the singular value decomposition

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{6}}{3} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} & \frac{1}{\sqrt{3}} \\ \frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix} \quad (4)$$

$A = U \quad \Sigma \quad V^T$

Since A has rank 2 (verify), it follows from (1) with $k = 2$ that the reduced singular value decomposition of A corresponding to (4) is

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{6}}{3} & 0 \\ \frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}$$

This yields the reduced singular value expansion

$$\begin{aligned} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} &= \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T = \sqrt{3} \begin{bmatrix} \frac{\sqrt{6}}{3} \\ \frac{\sqrt{6}}{6} \\ \frac{\sqrt{6}}{6} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} + (1) \begin{bmatrix} 0 \\ -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix} \\ &= \sqrt{3} \begin{bmatrix} \frac{\sqrt{3}}{3} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{3}}{6} & \frac{\sqrt{3}}{6} \\ \frac{\sqrt{3}}{6} & \frac{\sqrt{3}}{6} \end{bmatrix} + (1) \begin{bmatrix} 0 & 0 \\ -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \end{aligned}$$

Note that the matrices in the expansion have rank 1, as expected.

Data Compression and Image Processing

Singular value decompositions can be used to “compress” visual information for the purpose of reducing its required storage space and speeding up its electronic transmission. The first step in compressing a visual image is to represent it as a numerical matrix from which the visual image can be recovered when needed.

For example, a black and white photograph might be scanned as a rectangular array of pixels (points) and then stored as a matrix A by assigning each pixel a numerical value in accordance with its gray level. If 256 different gray levels are used (0 = white to 255 = black), then the entries in the matrix would be integers between 0 and 255. The image can be recovered from the matrix A by printing or displaying the pixels with their assigned gray levels.

If the matrix A has size $m \times n$, then one might store each of its mn entries individually. An alternative procedure is to compute the reduced singular value decomposition

$$A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \cdots + \sigma_k \mathbf{u}_k \mathbf{v}_k^T \quad (5)$$

in which $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k$, and store the σ 's, the $\mathbf{u}$'s, and the $\mathbf{v}$'s. When needed, the matrix A (and hence the image it represents) can be reconstructed from (5). Since each $\mathbf{u}_j$ has m entries and each $\mathbf{v}_j$ has n entries, this method requires storage space for

$$km + kn + k = k(m + n + 1)$$

numbers. Suppose, however, that the singular values $\sigma_{r+1}, \dots, \sigma_k$ are sufficiently small that dropping the corresponding terms in (5) produces an acceptable approximation

$$A_r = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \cdots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T \quad (6)$$

Historical Note



Original



Reconstruction

In 1924 the U.S. Federal Bureau of Investigation (FBI) began collecting fingerprints and handprints and now has more than 100 million such prints in its files. To reduce the storage cost, the FBI began working with the Los Alamos National Laboratory, the National Bureau of Standards, and other groups in 1993 to devise rank-based compression methods for storing prints in digital form. The adjacent figure shows an original fingerprint and a reconstruction from digital data that was compressed at a ratio of 26:1.

to A and the image that it represents. We call (6) the **rank r approximation of A** . This matrix requires storage space for only

$$rm + rn + r = r(m + n + 1)$$

numbers, compared to mn numbers required for entry-by-entry storage of A . For example, the rank 100 approximation of a 1000×1000 matrix A requires storage for only

$$100(1000 + 1000 + 1) = 200,100$$

numbers, compared to the 1,000,000 numbers required for entry-by-entry storage of A —a compression of almost 80%.

Figure 9.5.1 shows some approximations of a digitized mandrill image obtained using (6).

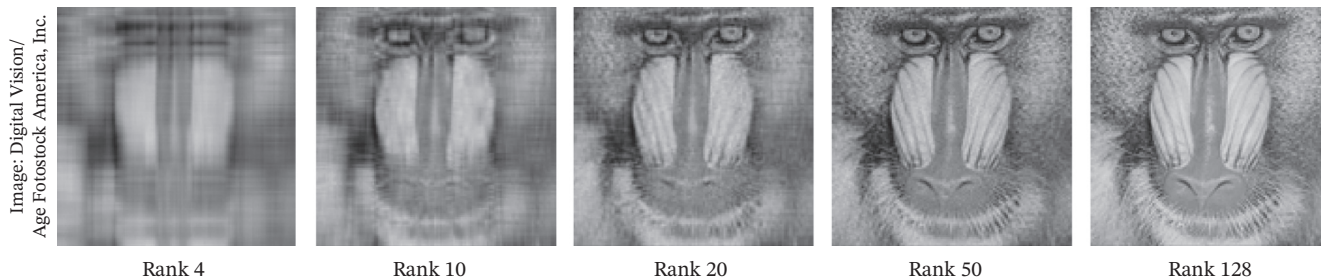


FIGURE 9.5.1

Exercise Set 9.5

In Exercises 1–4, find a reduced singular value decomposition of A . [Note: Each matrix appears in Exercise Set 9.4, where you were asked to find its (unreduced) singular value decomposition.]

1. $A = \begin{bmatrix} -2 & 2 \\ -1 & 1 \\ 2 & -2 \end{bmatrix}$
2. $A = \begin{bmatrix} -2 & -1 & 2 \\ 2 & 1 & -2 \end{bmatrix}$
3. $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ -1 & 1 \end{bmatrix}$
4. $A = \begin{bmatrix} 6 & 4 \\ 0 & 0 \\ 4 & 0 \end{bmatrix}$

In Exercises 5–8, find a reduced singular value expansion of A .

5. The matrix A in Exercise 1.
6. The matrix A in Exercise 2.
7. The matrix A in Exercise 3.

8. The matrix A in Exercise 4.
9. Suppose A is a 200×500 matrix. How many numbers must be stored in the rank 100 approximation of A ? Compare this with the number of entries of A .

True-False Exercises

- TF.** In parts (a)–(c) determine whether the statement is true or false, and justify your answer. Assume that $U_1 \Sigma_1 V_1^T$ is a reduced singular value decomposition of an $m \times n$ matrix of rank k .
- a. U_1 has size $m \times k$.
 - b. Σ_1 has size $k \times k$.
 - c. V_1 has size $k \times n$.

Chapter 9 Supplementary Exercises

1. Find an LU -decomposition of $A = \begin{bmatrix} -6 & 2 \\ 6 & 0 \end{bmatrix}$.
2. Find the LDU -decomposition of the matrix A in Exercise 1.
3. Find an LU -decomposition of $A = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 4 & 7 \\ 1 & 3 & 7 \end{bmatrix}$.
4. Find the LDU -decomposition of the matrix A in Exercise 3.
5. Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.
 - a. Identify the dominant eigenvalue of A and then find the corresponding dominant unit eigenvector $\mathbf{v}$ with positive entries.

- b. Apply the power method with Euclidean scaling to A and $\mathbf{x}_0$, stopping at $\mathbf{x}_5$. Compare your value of $\mathbf{x}_5$ to the eigenvector $\mathbf{v}$ found in part (a).
- c. Apply the power method with maximum entry scaling to A and $\mathbf{x}_0$, stopping at $\mathbf{x}_5$. Compare your result with the eigenvector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

6. Consider the symmetric matrix

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Discuss the behavior of the power sequence

$$\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_k, \dots$$

with Euclidean scaling for a general *nonzero* vector $\mathbf{x}_0$. What is it about the matrix that causes the observed behavior?

7. Suppose that a symmetric matrix A has distinct eigenvalues $\lambda_1 = 8$, $\lambda_2 = 1.4$, $\lambda_3 = 2.3$, and $\lambda_4 = -8.1$. What can you say about the convergence of the Rayleigh quotients?
8. Find a singular value decomposition of $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.
9. Find a singular value decomposition of $A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$.

10. Find a reduced singular value decomposition and a reduced singular value expansion of the matrix A in Exercise 9.
11. Find the reduced singular value decomposition of the matrix whose singular value decomposition is

$$A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 24 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & \frac{2}{3} \end{bmatrix}$$

12. Do orthogonally similar matrices have the same singular values? Justify your answer.
13. If P is the standard matrix for the orthogonal projection of R^n onto a subspace W , what can you say about the singular values of P ?
14. Prove: If A has rank 1, then there exists a scalar k such that $A^2 = kA$.

Applications of Linear Algebra

CHAPTER CONTENTS

10.1	Constructing Curves and Surfaces Through Specified Points	546
10.2	The Earliest Applications of Linear Algebra	551
10.3	Cubic Spline Interpolation	558
10.4	Markov Chains	568
10.5	Graph Theory	577
10.6	Games of Strategy	587
10.7	Forest Management	595
10.8	Computer Graphics	602
10.9	Equilibrium Temperature Distributions	610
10.10	Computed Tomography	619
10.11	Fractals	629
10.12	Chaos	645
10.13	Cryptography	658
10.14	Genetics	669
10.15	Age-Specific Population Growth	678
10.16	Harvesting of Animal Populations	687
10.17	A Least Squares Model for Human Hearing	695
10.18	Warps and Morphs	701
10.19	Internet Search Engines	710
10.20	Facial Recognition	716

Introduction

This chapter consists of 20 applications of linear algebra. With one clearly marked exception, each application is in its own independent section, so sections can be deleted or permuted as desired. Each topic begins with a list of linear algebra prerequisites.

Because our primary objective in this chapter is to present applications of linear algebra, proofs are often omitted. Whenever results from other fields are needed, they are stated precisely, with motivation where possible, but usually without proof.

10.1 Constructing Curves and Surfaces Through Specified Points

In this section we describe a technique that uses determinants to construct lines, circles, and general conic sections through specified points in the plane. The procedure is also used to pass planes and spheres in 3-space through fixed points.

PREREQUISITES: Linear Systems
Determinants
Analytic Geometry

The following theorem follows from Theorem 2.3.8.

Theorem 10.1.1

A homogeneous linear system with as many equations as unknowns has a nontrivial solution if and only if the determinant of the coefficient matrix is zero.

We will now show how this result can be used to determine equations of various curves and surfaces through specified points.

A Line Through Two Points

Suppose that (x_1, y_1) and (x_2, y_2) are two distinct points in the plane. There exists a unique line

$$c_1x + c_2y + c_3 = 0 \quad (1)$$

that passes through these two points (**Figure 10.1.1**). Note that c_1 , c_2 , and c_3 are not all zero and that these coefficients are unique only up to a multiplicative constant. Because (x_1, y_1) and (x_2, y_2) lie on the line, substituting them in (1) gives the two equations

$$c_1x_1 + c_2y_1 + c_3 = 0 \quad (2)$$

$$c_1x_2 + c_2y_2 + c_3 = 0 \quad (3)$$

The three equations, (1), (2), and (3), can be grouped together and rewritten as

$$xc_1 + yc_2 + c_3 = 0$$

$$x_1c_1 + y_1c_2 + c_3 = 0$$

$$x_2c_1 + y_2c_2 + c_3 = 0$$

which is a homogeneous linear system of three equations for c_1 , c_2 , and c_3 . Because c_1 , c_2 , and c_3 are not all zero, this system has a nontrivial solution, so the determinant of the coefficient matrix of the system must be zero. That is,

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0 \quad (4)$$

Consequently, every point (x, y) on the line satisfies (4); conversely, it can be shown that every point (x, y) that satisfies (4) lies on the line.

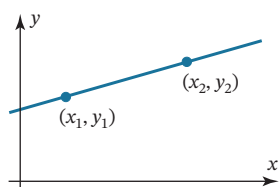


FIGURE 10.1.1

EXAMPLE 1 | Equation of a Line

Find the equation of the line that passes through the two points (2, 1) and (3, 7).

Solution Substituting the coordinates of the two points into Equation (4) gives

$$\begin{vmatrix} x & y & 1 \\ 2 & 1 & 1 \\ 3 & 7 & 1 \end{vmatrix} = 0$$

The cofactor expansion of this determinant along the first row then gives

$$-6x + y + 11 = 0$$

A Circle Through Three Points

Suppose that there are three distinct points in the plane, (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) , not all lying on a straight line. From analytic geometry we know that there is a unique circle, say,

$$c_1(x^2 + y^2) + c_2x + c_3y + c_4 = 0 \quad (5)$$

that passes through them (Figure 10.1.2). Substituting the coordinates of the three points into this equation gives

$$c_1(x_1^2 + y_1^2) + c_2x_1 + c_3y_1 + c_4 = 0 \quad (6)$$

$$c_1(x_2^2 + y_2^2) + c_2x_2 + c_3y_2 + c_4 = 0 \quad (7)$$

$$c_1(x_3^2 + y_3^2) + c_2x_3 + c_3y_3 + c_4 = 0 \quad (8)$$

As before, Equations (5) through (8) form a homogeneous linear system with a nontrivial solution for c_1 , c_2 , c_3 , and c_4 . Thus the determinant of the coefficient matrix is zero:

$$\begin{vmatrix} x^2 + y^2 & x & y & 1 \\ x_1^2 + y_1^2 & x_1 & y_1 & 1 \\ x_2^2 + y_2^2 & x_2 & y_2 & 1 \\ x_3^2 + y_3^2 & x_3 & y_3 & 1 \end{vmatrix} = 0 \quad (9)$$

This is a determinant form for the equation of the circle.

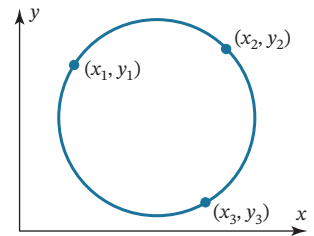


FIGURE 10.1.2

EXAMPLE 2 | Equation of a Circle

Find the equation of the circle that passes through the three points (1, 7), (6, 2), and (4, 6).

Solution Substituting the coordinates of the three points into Equation (9) gives

$$\begin{vmatrix} x^2 + y^2 & x & y & 1 \\ 50 & 1 & 7 & 1 \\ 40 & 6 & 2 & 1 \\ 52 & 4 & 6 & 1 \end{vmatrix} = 0$$

which reduces to

$$10(x^2 + y^2) - 20x - 40y - 200 = 0$$

In standard form this is

$$(x - 1)^2 + (y - 2)^2 = 5^2$$

Thus the circle has center (1, 2) and radius 5.

A General Conic Section Through Five Points

In his monumental work *Principia Mathematica*, Issac Newton posed and solved the following problem (Book I, Proposition 22, Problem 14): “To describe a conic that shall pass through five given points.” Newton solved this problem geometrically, as shown in **Figure 10.1.3**, in which he passed an ellipse through the points A, B, D, P, C; however, the methods of this section can also be applied.

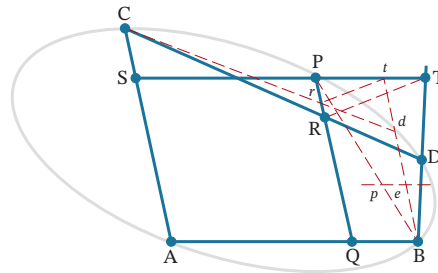


FIGURE 10.1.3

The general equation of a conic section in the plane (a parabola, hyperbola, or ellipse, or degenerate forms of these curves) is given by

$$c_1x^2 + c_2xy + c_3y^2 + c_4x + c_5y + c_6 = 0$$

This equation contains six coefficients, but we can reduce the number to five if we divide through by any one of them that is not zero. Thus only five coefficients must be determined, so five distinct points in the plane are sufficient to determine the equation of the conic section (**Figure 10.1.4**). As before, the equation can be put in determinant form (see Exercise 7):

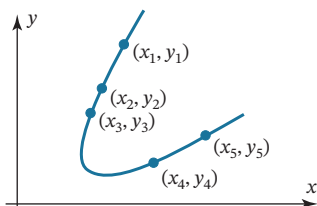


FIGURE 10.1.4

$$\begin{vmatrix} x^2 & xy & y^2 & x & y & 1 \\ x_1^2 & x_1y_1 & y_1^2 & x_1 & y_1 & 1 \\ x_2^2 & x_2y_2 & y_2^2 & x_2 & y_2 & 1 \\ x_3^2 & x_3y_3 & y_3^2 & x_3 & y_3 & 1 \\ x_4^2 & x_4y_4 & y_4^2 & x_4 & y_4 & 1 \\ x_5^2 & x_5y_5 & y_5^2 & x_5 & y_5 & 1 \end{vmatrix} = 0 \quad (10)$$

EXAMPLE 3 | Equation of an Orbit

An astronomer who wants to determine the orbit of an asteroid about the Sun sets up a Cartesian coordinate system in the plane of the orbit with the Sun at the origin. Astronomical units of measurement are used along the axes (1 astronomical unit = mean distance of Earth to Sun = 93 million miles). By Kepler's first law, the orbit must be an ellipse, so the astronomer makes five observations of the asteroid at five different times and finds five points along the orbit to be

$$(8.025, 8.310), (10.170, 6.355), (11.202, 3.212), (10.736, 0.375), (9.092, -2.267)$$

Find the equation of the orbit.

Solution Substituting the coordinates of the five given points into (10) and rounding to three decimal places give

$$\begin{vmatrix} x^2 & xy & y^2 & x & y & 1 \\ 64.401 & 66.688 & 69.056 & 8.025 & 8.310 & 1 \\ 103.429 & 64.630 & 40.386 & 10.170 & 6.355 & 1 \\ 125.485 & 35.981 & 10.317 & 11.202 & 3.212 & 1 \\ 115.262 & 4.026 & 0.141 & 10.736 & 0.375 & 1 \\ 82.664 & -20.612 & 5.139 & 9.092 & -2.267 & 1 \end{vmatrix} = 0$$

The cofactor expansion of this determinant along the first row yields

$$386.802x^2 - 102.895xy + 446.029y^2 - 2476.443x - 1427.998y - 17109.375 = 0$$

Figure 10.1.5 is an accurate diagram of the orbit, together with the five given points.

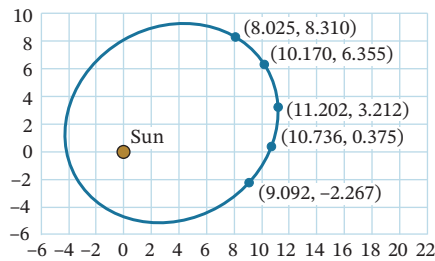


FIGURE 10.1.5

A Plane Through Three Points

In Exercise 8 we ask you to show the following: The plane in 3-space with equation

$$c_1x + c_2y + c_3z + c_4 = 0$$

that passes through three noncollinear points (x_1, y_1, z_1) , (x_2, y_2, z_2) , and (x_3, y_3, z_3) is given by the determinant equation

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0 \quad (11)$$

EXAMPLE 4 | Equation of a Plane

The equation of the plane that passes through the three noncollinear points $(1, 1, 0)$, $(2, 0, -1)$, and $(2, 9, 2)$ is

$$\begin{vmatrix} x & y & z & 1 \\ 1 & 1 & 0 & 1 \\ 2 & 0 & -1 & 1 \\ 2 & 9 & 2 & 1 \end{vmatrix} = 0$$

which reduces to

$$2x - y + 3z - 1 = 0$$

A Sphere Through Four Points

In Exercise 9 we ask you to show the following: The sphere in 3-space with equation

$$c_1(x^2 + y^2 + z^2) + c_2x + c_3y + c_4z + c_5 = 0$$

that passes through four noncoplanar points (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) , and (x_4, y_4, z_4) is given by the following determinant equation:

$$\begin{vmatrix} x^2 + y^2 + z^2 & x & y & z & 1 \\ x_1^2 + y_1^2 + z_1^2 & x_1 & y_1 & z_1 & 1 \\ x_2^2 + y_2^2 + z_2^2 & x_2 & y_2 & z_2 & 1 \\ x_3^2 + y_3^2 + z_3^2 & x_3 & y_3 & z_3 & 1 \\ x_4^2 + y_4^2 + z_4^2 & x_4 & y_4 & z_4 & 1 \end{vmatrix} = 0 \quad (12)$$

EXAMPLE 5 | Equation of a Sphere

The equation of the sphere that passes through the four points $(0, 3, 2)$, $(1, -1, 1)$, $(2, 1, 0)$, and $(5, 1, 3)$ is

$$\begin{vmatrix} x^2 + y^2 + z^2 & x & y & z & 1 \\ 13 & 0 & 3 & 2 & 1 \\ 3 & 1 & -1 & 1 & 1 \\ 5 & 2 & 1 & 0 & 1 \\ 35 & 5 & 1 & 3 & 1 \end{vmatrix} = 0$$

This reduces to

$$x^2 + y^2 + z^2 - 4x - 2y - 6z + 5 = 0$$

which in standard form is

$$(x - 2)^2 + (y - 1)^2 + (z - 3)^2 = 9$$

Exercise Set 10.1

1. Find the equations of the lines that pass through the following points:
 - a. $(1, -1)$, $(2, 2)$
 - b. $(0, 1)$, $(1, -1)$
2. Find the equations of the circles that pass through the following points:
 - a. $(2, 6)$, $(2, 0)$, $(5, 3)$
 - b. $(2, -2)$, $(3, 5)$, $(-4, 6)$
3. Find the equation of the conic section that passes through the points $(0, 0)$, $(0, -1)$, $(2, 0)$, $(2, -5)$, and $(4, -1)$.
4. Find the equations of the planes in 3-space that pass through the following points:
 - a. $(1, 1, -3)$, $(1, -1, 1)$, $(0, -1, 2)$
 - b. $(2, 3, 1)$, $(2, -1, -1)$, $(1, 2, 1)$
5. a. Alter Equation (11) so that it determines the plane that passes through the origin and is parallel to the plane that passes through three specified noncollinear points.
 b. Find the two planes described in part (a) corresponding to the triplets of points in Exercises 4(a) and 4(b).
6. Find the equations of the spheres in 3-space that pass through the following points:
 - a. $(1, 2, 3)$, $(-1, 2, 1)$, $(1, 0, 1)$, $(1, 2, -1)$
 - b. $(0, 1, -2)$, $(1, 3, 1)$, $(2, -1, 0)$, $(3, 1, -1)$
7. Show that Equation (10) is the equation of the conic section that passes through five given distinct points in the plane.
8. Show that Equation (11) is the equation of the plane in 3-space that passes through three given noncollinear points.
9. Show that Equation (12) is the equation of the sphere in 3-space that passes through four given noncoplanar points.
10. Find a determinant equation for the parabola of the form

$$c_1y + c_2x^2 + c_3x + c_4 = 0$$
 that passes through three given noncollinear points in the plane.
11. What does Equation (9) become if the three distinct points are collinear?

12. What does Equation (11) become if the three distinct points are collinear?
13. What does Equation (12) become if the four points are coplanar?

Working with Technology

The following exercises are designed to be solved using a technology utility. Typically, this will be MATLAB, *Mathematica*, Maple, Derive, or Mathcad, but it may also be some other type of linear algebra software or a scientific calculator with some linear algebra capabilities. For each exercise you will need to read the relevant documentation for the particular utility you are using. The goal of these exercises is to provide you with a basic proficiency with your technology utility. Once you have mastered the techniques in these exercises, you will be able to use your technology utility to solve many of the problems in the regular exercise sets.

T1. The general equation of a quadric surface is given by

$$a_1x^2 + a_2y^2 + a_3z^2 + a_4xy + a_5xz + a_6yz + a_7x + a_8y + a_9z + a_{10} = 0$$

Given nine points on this surface, it may be possible to determine its equation.

- a.** Show that if the nine points (x_i, y_i) for $i = 1, 2, 3, \dots, 9$ lie on this surface, and if they determine uniquely the equation of this surface, then its equation can be written in determinant form as

$$\begin{vmatrix} x^2 & y^2 & z^2 & xy & xz & yz & x & y & z & 1 \\ x_1^2 & y_1^2 & z_1^2 & x_1y_1 & x_1z_1 & y_1z_1 & x_1 & y_1 & z_1 & 1 \\ x_2^2 & y_2^2 & z_2^2 & x_2y_2 & x_2z_2 & y_2z_2 & x_2 & y_2 & z_2 & 1 \\ x_3^2 & y_3^2 & z_3^2 & x_3y_3 & x_3z_3 & y_3z_3 & x_3 & y_3 & z_3 & 1 \\ x_4^2 & y_4^2 & z_4^2 & x_4y_4 & x_4z_4 & y_4z_4 & x_4 & y_4 & z_4 & 1 \\ x_5^2 & y_5^2 & z_5^2 & x_5y_5 & x_5z_5 & y_5z_5 & x_5 & y_5 & z_5 & 1 \\ x_6^2 & y_6^2 & z_6^2 & x_6y_6 & x_6z_6 & y_6z_6 & x_6 & y_6 & z_6 & 1 \\ x_7^2 & y_7^2 & z_7^2 & x_7y_7 & x_7z_7 & y_7z_7 & x_7 & y_7 & z_7 & 1 \\ x_8^2 & y_8^2 & z_8^2 & x_8y_8 & x_8z_8 & y_8z_8 & x_8 & y_8 & z_8 & 1 \\ x_9^2 & y_9^2 & z_9^2 & x_9y_9 & x_9z_9 & y_9z_9 & x_9 & y_9 & z_9 & 1 \end{vmatrix} = 0$$

- b.** Use the result in part (a) to determine the equation of the quadric surface that passes through the points $(1, 2, 3)$, $(2, 1, 7)$, $(0, 4, 6)$, $(3, -1, 4)$, $(3, 0, 11)$, $(-1, 5, 8)$, $(9, -8, 3)$, $(4, 5, 3)$, and $(-2, 6, 10)$.

T2. a. A hyperplane in the n -dimensional Euclidean space R^n has an equation of the form

$$a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n + a_{n+1} = 0$$

where $a_i, i = 1, 2, 3, \dots, n+1$, are constants, not all zero, and $x_i, i = 1, 2, 3, \dots, n$, are variables for which

$$(x_1, x_2, x_3, \dots, x_n) \in R^n$$

A point

$$(x_{10}, x_{20}, x_{30}, \dots, x_{n0}) \in R^n$$

lies on this hyperplane if

$$a_1x_{10} + a_2x_{20} + a_3x_{30} + \cdots + a_nx_{n0} + a_{n+1} = 0$$

Given that the n points $(x_{1i}, x_{2i}, x_{3i}, \dots, x_{ni})$, $i = 1, 2, 3, \dots, n$, lie on this hyperplane and that they uniquely determine the equation of the hyperplane, show that the equation of the hyperplane can be written in determinant form as

$$\begin{vmatrix} x_1 & x_2 & x_3 & \cdots & x_n & 1 \\ x_{11} & x_{21} & x_{31} & \cdots & x_{n1} & 1 \\ x_{12} & x_{22} & x_{32} & \cdots & x_{n2} & 1 \\ x_{13} & x_{23} & x_{33} & \cdots & x_{n3} & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{1n} & x_{2n} & x_{3n} & \cdots & x_{nn} & 1 \end{vmatrix} = 0$$

- b.** Determine the equation of the hyperplane in R^9 that goes through the following nine points:

$$\begin{array}{ll} (1, 2, 3, 4, 5, 6, 7, 8, 9) & (2, 3, 4, 5, 6, 7, 8, 9, 1) \\ (3, 4, 5, 6, 7, 8, 9, 1, 2) & (4, 5, 6, 7, 8, 9, 1, 2, 3) \\ (5, 6, 7, 8, 9, 1, 2, 3, 4) & (6, 7, 8, 9, 1, 2, 3, 4, 5) \\ (7, 8, 9, 1, 2, 3, 4, 5, 6) & (8, 9, 1, 2, 3, 4, 5, 6, 7) \\ (9, 1, 2, 3, 4, 5, 6, 7, 8) & \end{array}$$

10.2 The Earliest Applications of Linear Algebra

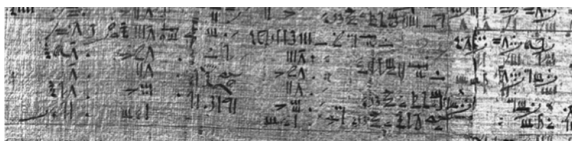
Linear systems can be found in the earliest writings of many ancient civilizations. In this section we give some examples of the types of problems that they used to solve.

PREREQUISITES: Linear Systems

The practical problems of early civilizations included the measurement of land, the distribution of goods, the tracking of resources such as wheat and cattle, and taxation and inheritance calculations. In many cases, these problems led to linear systems of equations since

linearity is one of the simplest relationships that can exist among variables. In this section we present examples from five diverse ancient cultures illustrating how they used and solved systems of linear equations. We restrict ourselves to examples before A.D. 500. These examples consequently predate the development of the field of algebra by Islamic/Arab mathematicians, a field that ultimately led in the nineteenth century to the branch of mathematics now called linear algebra.

EXAMPLE 1 | Egypt (about 1650 B.C.)



Problem 40 of the Ahmes Papyrus

[Image: © The Trustees of the British Museum]

The Ahmes (or Rhind) Papyrus is the source of most of our information about ancient Egyptian mathematics. This 5-meter-long papyrus contains 84 short mathematical problems, together with their solutions, and dates from about 1650 B.C. Problem 40 in this papyrus is the following:

Divide 100 hekats of barley among five men in arithmetic progression so that the sum of the two smallest is one-seventh the sum of the three largest.

Let a be the least amount that any man obtains, and let d be the common difference of the terms in the arithmetic progression. Then the other four men receive $a + d$, $a + 2d$, $a + 3d$, and $a + 4d$ hekats. The two conditions of the problem require that

$$\begin{aligned} a + (a + d) + (a + 2d) + (a + 3d) + (a + 4d) &= 100 \\ \frac{1}{7}[(a + 2d) + (a + 3d) + (a + 4d)] &= a + (a + d) \end{aligned}$$

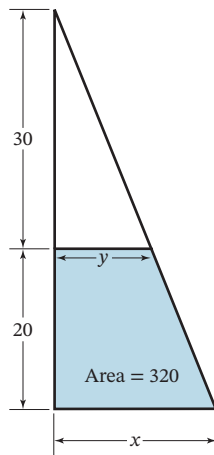
These equations reduce to the following system of two equations in two unknowns:

$$\begin{aligned} 5a + 10d &= 100 \\ 11a - 2d &= 0 \end{aligned} \tag{1}$$

The solution technique described in the papyrus is known as the method of false position or false assumption. It begins by assuming some convenient value of a (in our case $a = 1$), substituting that value into the second equation, and obtaining $d = 11/2$. Substituting $a = 1$ and $d = 11/2$ into the left-hand side of the first equation gives 60, whereas the right-hand side is 100. Adjusting the initial guess for a by multiplying it by $100/60$ leads to the correct value $a = 5/3$. Substituting $a = 5/3$ into the second equation then gives $d = 55/6$, so the quantities of barley received by the five men are $10/6$, $65/6$, $120/6$, $175/6$, and $230/6$ hekats. This technique of guessing a value of an unknown and later adjusting it has been used by many cultures throughout the ages.

EXAMPLE 2 | Babylonia (1900–1600 B.C.)

The Old Babylonian Empire flourished in Mesopotamia between 1900 and 1600 B.C. Many clay tablets containing mathematical tables and problems survive from that period, one of which (designated Ca MLA 1950) contains the next problem. The statement of the problem is a bit muddled because of the condition of the tablet, but the diagram and the solution on the tablet indicate that the problem is as follows:

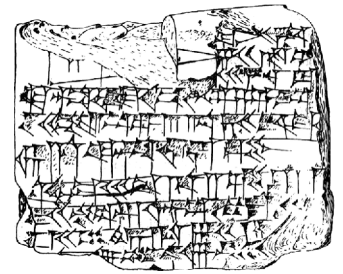


A trapezoid with an area of 320 square units is cut off from a right triangle by a line parallel to one of its sides. The other side has length 50 units, and the height of the trapezoid is 20 units. What are the upper and the lower widths of the trapezoid?

Let x be the lower width of the trapezoid and y its upper width. The area of the trapezoid is its height times its average width, so $20\left(\frac{x+y}{2}\right) = 320$. Using similar triangles, we also have $\frac{x}{50} = \frac{y}{30}$. The solution on the tablet uses these relations to generate the linear system

$$\begin{aligned}\frac{1}{2}(x+y) &= 16 \\ \frac{1}{2}(x-y) &= 4\end{aligned}\tag{2}$$

Adding and subtracting these two equations then gives the solution $x = 20$ and $y = 12$.



Babylonian clay tablet Ca MLA 1950 [Image: American Oriental Society/American Schools of Oriental Research]

EXAMPLE 3 | China (A.D. 263)

The most important treatise in the history of Chinese mathematics is the Chiu Chang Suan Shu, or “The Nine Chapters of the Mathematical Art.” This treatise, which is a collection of 246 problems and their solutions, was assembled in its final form by Liu Hui in A.D. 263. Its contents, however, go back to at least the beginning of the Han dynasty in the second century B.C. The eighth of its nine chapters, entitled “The Way of Calculating by Arrays,” contains 18 word problems that lead to linear systems in three to six unknowns. The general solution procedure described is almost identical to the Gaussian elimination technique developed in Europe in the nineteenth century by Carl Friedrich Gauss (see page 16). The first problem in the eighth chapter is the following:

九章算術

Chiu Chang Suan Shu in Chinese characters

There are three classes of corn, of which three bundles of the first class, two of the second, and one of the third make 39 measures. Two of the first, three of the second, and one of the third make 34 measures. And one of the first, two of the second, and three of the third make 26 measures. How many measures of grain are contained in one bundle of each class?

Let x , y , and z be the measures of the first, second, and third classes of corn. Then the conditions of the problem lead to the following linear system of three equations in three unknowns:

$$\begin{aligned} 3x + 2y + z &= 39 \\ 2x + 3y + z &= 34 \\ x + 2y + 3z &= 26 \end{aligned} \quad (3)$$

The solution described in the treatise represented the coefficients of each equation by an appropriate number of rods placed within squares on a counting table. Positive coefficients were represented by black rods, negative coefficients were represented by red rods, and the squares corresponding to zero coefficients were left empty. The counting table was laid out as follows so that the coefficients of each equation appear in columns with the first equation in the rightmost column:

1	2	3
2	3	2
3	1	1
26	34	39

Next, the numbers of rods within the squares were adjusted to accomplish the following two steps: (1) two times the numbers of the third column were subtracted from three times the numbers in the second column and (2) the numbers in the third column were subtracted from three times the numbers in the first column. The result was the following array:

		3
4	5	2
8	1	1
39	24	39

In this array, four times the numbers in the second column were subtracted from five times the numbers in the first column, yielding

		3
	5	2
36	1	1
99	24	39

This last array is equivalent to the linear system

$$\begin{aligned} 3x + 2y + z &= 39 \\ 5y + z &= 24 \\ 36z &= 99 \end{aligned}$$

This triangular system was solved by a method equivalent to back substitution to obtain $x = 37/4$, $y = 17/4$, and $z = 11/4$.

EXAMPLE 4 | Greece (third century B.C.)

Perhaps the most famous system of linear equations from antiquity is the one associated with the first part of Archimedes' celebrated Cattle Problem. This problem supposedly was posed by Archimedes as a challenge to his colleague Eratosthenes. No solution has come down to us from ancient times, so that it is not known how, or even whether, either of these two geometers solved it.

If thou art diligent and wise, O stranger, compute the number of cattle of the Sun, who once upon a time grazed on the fields of the Thrinacian isle of Sicily, divided into four herds of different colors, one milk white, another glossy black, a third yellow, and the last dappled. In each herd were bulls, mighty in number according to these proportions: Understand, stranger, that the white bulls were equal to a half and a third of the black together with the whole of the yellow, while the black were equal to the fourth part of the dappled and a fifth, together with, once more, the whole of the yellow. Observe further that the remaining bulls, the dappled, were equal to a sixth part of the white and a seventh, together with all of the yellow. These were the proportions of the cows: The white were precisely equal to the third part and a fourth of the whole herd of the black; while the black were equal to the fourth part once more of the dappled and with it a fifth part, when all, including the bulls, went to pasture together. Now the dappled in four parts were equal in number to a fifth part and a sixth of the yellow herd. Finally the yellow were in number equal to a sixth part and a seventh of the white herd. If thou canst accurately tell, O stranger, the number of cattle of the Sun, giving separately the number of well-fed bulls and again the number of females according to each color, thou wouldst not be called unskilled or ignorant of numbers, but not yet shalt thou be numbered among the wise.



Archimedes c. 287–212 B.C.

The conventional designation of the eight variables in this problem is

- W = number of white bulls
- B = number of black bulls
- Y = number of yellow bulls
- D = number of dappled bulls
- w = number of white cows
- b = number of black cows
- y = number of yellow cows
- d = number of dappled cows

The problem can now be stated as the following seven homogeneous equations in eight unknowns:

1. $W = \left(\frac{1}{2} + \frac{1}{3}\right)B + Y$ (The white bulls were equal to a half and a third of the black [bulls] together with the whole of the yellow [bulls].)
2. $B = \left(\frac{1}{4} + \frac{1}{5}\right)D + Y$ (The black [bulls] were equal to the fourth part of the dappled [bulls] and a fifth, together with, once more, the whole of the yellow [bulls].)
3. $D = \left(\frac{1}{6} + \frac{1}{7}\right)W + Y$ (The remaining bulls, the dappled, were equal to a sixth part of the white [bulls] and a seventh, together with all of the yellow [bulls].)
4. $w = \left(\frac{1}{3} + \frac{1}{4}\right)(B + b)$ (The white [cows] were precisely equal to the third part and a fourth of the whole herd of the black.)

5. $b = \left(\frac{1}{4} + \frac{1}{5}\right)(D + d)$ (The black [cows] were equal to the fourth part once more of the dappled and with it a fifth part, when all, including the bulls, went to pasture together.)
6. $d = \left(\frac{1}{5} + \frac{1}{6}\right)(Y + y)$ (The dappled [cows] in four parts [that is, in totality] were equal in number to a fifth part and a sixth of the yellow herd.)
7. $y = \left(\frac{1}{6} + \frac{1}{7}\right)(W + w)$ (The yellow [cows] were in number equal to a sixth part and a seventh of the white herd.)

As we ask you to show in the exercises, this system has infinitely many solutions of the form

$$\begin{aligned}
 W &= 10,366,482k \\
 B &= 7,460,514k \\
 Y &= 4,149,387k \\
 D &= 7,358,060k \\
 w &= 7,206,360k \\
 b &= 4,893,246k \\
 y &= 5,439,213k \\
 d &= 3,515,820k
 \end{aligned} \tag{4}$$

where k is any real number. The values $k = 1, 2, \dots$ give infinitely many positive integer solutions to the problem, with $k = 1$ giving the smallest solution.



Fragment III-5-3v of the Bakhshali Manuscript

[Image: Bodleian Library, University of Oxford, MS. Sansk. d. 14, fragment III 5 3v.]

EXAMPLE 5 | India (fourth century A.D.)

The Bakhshali Manuscript is an ancient work of Indian/Hindu mathematics dating from around the fourth century A.D., although some of its materials undoubtedly come from many centuries before. It consists of about 70 leaves or sheets of birch bark containing mathematical problems and their solutions. Many of its problems are so-called equalization problems that lead to systems of linear equations. One such problem on the fragment shown is the following:

One merchant has seven asava horses, a second has nine haya horses, and a third has ten camels. They are equally well off in the value of their animals if each gives two animals, one to each of the others. Find the price of each animal and the total value of the animals possessed by each merchant.

Let x be the price of an asava horse, let y be the price of a haya horse, let z be the price of a camel, and let K be the total value of the animals possessed by each merchant. Then the conditions of the problem lead to the following system of equations:

$$\begin{aligned}
 5x + y + z &= K \\
 x + 7y + z &= K \\
 x + y + 8z &= K
 \end{aligned} \tag{5}$$

The method of solution described in the manuscript begins by subtracting the quantity $(x + y + z)$ from both sides of the three equations to obtain $4x = 6y = 7z = K - (x + y + z)$. This shows that if the prices x , y , and z are to be integers, then the quantity $K - (x + y + z)$ must be an integer that is divisible by 4, 6, and 7. The manuscript takes the product of these three numbers, or 168, for the value of $K - (x + y + z)$, which yields $x = 42$, $y = 28$, and $z = 24$ for the prices and $K = 262$ for the total value. (See Exercise 6 for more solutions to this problem.)

Exercise Set 10.2

1. The following lines from Book 12 of Homer's *Odyssey* relate a precursor of Archimedes' Cattle Problem:

Thou shalt ascend the isle triangular,
Where many oxen of the Sun are fed,
And fatted flocks. Of oxen fifty head
In every herd feed, and their herds are seven;
And of his fat flocks is their number even.

The last line means that there are as many sheep in all the flocks as there are oxen in all the herds. What is the total number of oxen and sheep that belong to the god of the Sun? (This was a difficult problem in Homer's day.)

2. Solve the following problems from the Bakhshali Manuscript.

- a. B possesses two times as much as A; C has three times as much as A and B together; D has four times as much as A, B, and C together. Their total possessions are 300. What is the possession of A?
- b. B gives 2 times as much as A; C gives 3 times as much as B; D gives 4 times as much as C. Their total gift is 132. What is the gift of A?

3. A problem on a Babylonian tablet requires finding the length and width of a rectangle given that the length and the width add up to 10, while the length and one-fourth of the width add up to 7. The solution provided on the tablet consists of the following four statements:

Multiply 7 by 4 to obtain 28.
Take away 10 from 28 to obtain 18.
Take one-third of 18 to obtain 6, the length.
Take away 6 from 10 to obtain 4, the width.

Explain how these steps lead to the answer.

4. The following two problems are from "The Nine Chapters of the Mathematical Art." Solve them using the array technique described in Example 3.

- a. Five oxen and two sheep are worth 10 units and two oxen and five sheep are worth 8 units. What is the value of each ox and sheep?
- b. There are three kinds of corn. The grains contained in two, three, and four bundles, respectively, of these three classes of corn, are not sufficient to make a whole measure. However, if we added to them one bundle of the second, third, and first classes, respectively, then the grains would become on full measure in each case. How many measures of grain does each bundle of the different classes contain?

5. This problem in part (a) is known as the "Flower of Thymaridas," named after a Pythagorean of the fourth century B.C.

- a. Given the n numbers $a_1, a_2, \dots, a_n$, solve for $x_1, x_2, \dots, x_n$ in the following linear system:

$$\begin{aligned} x_1 + x_2 + \cdots + x_n &= a_1 \\ x_1 + x_2 &= a_2 \\ x_1 + x_3 &= a_3 \\ &\vdots \\ x_1 + x_n &= a_n \end{aligned}$$

- b. Identify a problem in this exercise set that fits the pattern in part (a), and solve it using your general solution.

6. For Example 5 from the Bakhshali Manuscript:

- a. Express Equations (5) as a homogeneous linear system of three equations in four unknowns (x, y, z , and K) and show that the solution set has one arbitrary parameter.
- b. Find the smallest solution for which all four variables are positive integers.
- c. Show that the solution given in Example 5 is included among your solutions.

7. Solve the problems posed in the following three epigrams, which appear in a collection entitled "The Greek Anthology," compiled in part by a scholar named Metrodorus around A.D. 500. Some of its 46 mathematical problems are believed to date as far back as 600 B.C. [Note: Before solving parts (a) and (c), you will have to formulate the question.]

- a. I desire my two sons to receive the thousand staters of which I am possessed, but let the fifth part of the legitimate one's share exceed by ten the fourth part of what falls to the illegitimate one.
- b. Make me a crown weighing sixty minae, mixing gold and brass, and with them tin and much-wrought iron. Let the gold and brass together form two-thirds, the gold and tin together three-fourths, and the gold and iron three-fifths. Tell me how much gold you must put in, how much brass, how much tin, and how much iron, so as to make the whole crown weigh sixty minae.
- c. First person: I have what the second has and the third of what the third has. Second person: I have what the third has and the third of what the first has. Third person: And I have ten minae and the third of what the second has.

Working with Technology

The following exercises are designed to be solved using a technology utility. Typically, this will be MATLAB, *Mathematica*, Maple, Derive, or Mathcad, but it may also be some other type of linear algebra software or a scientific calculator with some linear algebra capabilities. For each exercise you will need to read the relevant documentation for the particular utility you are using. The goal of these exercises is to provide you with a basic proficiency with your technology utility. Once you have mastered the techniques in these exercises, you will be able to use your technology utility to solve many of the problems in the regular exercise sets.

- T1. a. Solve Archimedes' Cattle Problem using a symbolic algebra program.

- b. The Cattle Problem has a second part in which two additional conditions are imposed. The first of these states that "When the white bulls mingled their number with the black, they stood firm, equal in depth and breadth." This requires that $W + B$ be a square number, that is, 1, 4, 9, 16, 25, and so on. Show that this requires that the values of k in Eq. (4) be restricted as follows:

$$k = 4,456,749r^2, \quad r = 1, 2, 3, \dots$$

and find the smallest total number of cattle that satisfies this second condition.

Remark The second condition imposed in the second part of the Cattle Problem states that “When the yellow and the dappled bulls were gathered into one herd, they stood in such a manner that their number, beginning from one, grew slowly greater ’til it completed a triangular figure.” This requires that the quantity $Y + D$ be a triangular number—that is, a number of the form $1, 1 + 2, 1 + 2 + 3, 1 + 2 + 3 + 4, \dots$. This final part of the problem was not completely solved until 1965 when all 206,545 digits of the smallest number of cattle that satisfies this condition were found using a computer.

T2. The following problem is from “The Nine Chapters of the Mathematical Art” and determines a homogeneous linear

system of five equations in six unknowns. Show that the system has infinitely many solutions, and find the one for which the depth of the well and the lengths of the five ropes are the smallest possible positive integers.

Suppose that five families share a well. Suppose further that

2 of A’s ropes are short of the well’s depth by one of B’s ropes.

3 of B’s ropes are short of the well’s depth by one of C’s ropes.

4 of C’s ropes are short of the well’s depth by one of D’s ropes.

5 of D’s ropes are short of the well’s depth by one of E’s ropes.

6 of E’s ropes are short of the well’s depth by one of A’s ropes.

10.3 Cubic Spline Interpolation

In this section an artist’s drafting aid is used as a physical model for the mathematical problem of finding a curve that passes through specified points in the plane. The parameters of the curve are determined by solving a linear system of equations.

PREREQUISITES: Linear Systems
Matrix Algebra
Differential Calculus

Curve Fitting

Fitting a curve through specified points in the plane is a common problem encountered in analyzing experimental data, in ascertaining the relations among variables, and in design work. A ubiquitous application is in the design and description of computer and printer fonts, such as PostScript™ and TrueType™ fonts (**Figure 10.3.1**). In **Figure 10.3.2** seven points in the xy -plane are displayed, and in **Figure 10.3.4** a smooth curve has been drawn that passes through them. A curve that passes through a set of points in the plane is said to **interpolate** those points, and the curve is called an **interpolating curve** for those points.

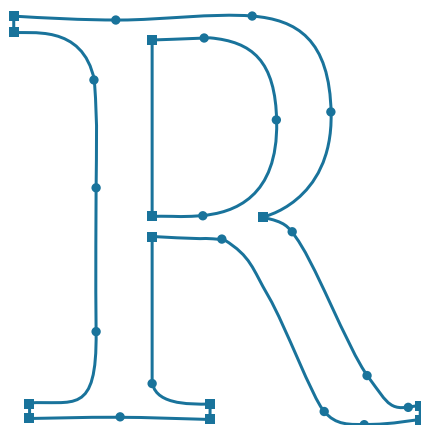


FIGURE 10.3.1

The interpolating curve in Figure 10.3.4 was drawn with the aid of a *drafting spline* (Figure 10.3.3). This drafting aid consists of a thin, flexible strip of wood or other material that is bent to pass through the points to be interpolated. Attached sliding weights hold the spline in position while the artist draws the interpolating curve. The drafting spline will serve as the physical model for a mathematical theory of interpolation that we will discuss in this section.

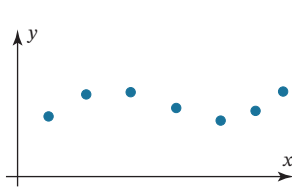


FIGURE 10.3.2

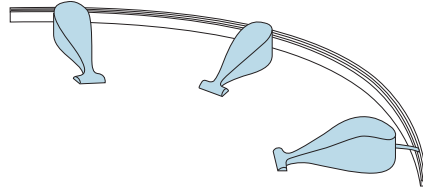


FIGURE 10.3.3

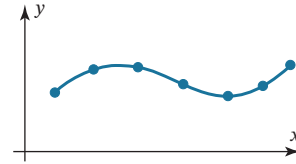


FIGURE 10.3.4

Statement of the Problem

Suppose that we are given n points in the xy -plane,

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

which we wish to interpolate with a “well-behaved” curve (Figure 10.3.5). For convenience, we take the points to be equally spaced in the x -direction, although our results can easily be extended to the case of unequally spaced points. If we let the common distance between the x -coordinates of the points be h , then we have

$$x_2 - x_1 = x_3 - x_2 = \dots = x_n - x_{n-1} = h$$

Let $y = S(x)$, $x_1 \leq x \leq x_n$ denote the interpolating curve that we seek. We assume that this curve describes the displacement of a drafting spline that interpolates the n points when the weights holding down the spline are situated precisely at the n points. It is known from linear beam theory that for small displacements, the fourth derivative of the displacement of a beam is zero along any interval of the x -axis that contains no external forces acting on the beam. If we treat our drafting spline as a thin beam and realize that the only external forces acting on it arise from the weights at the n specified points, then it follows that

$$S^{(iv)}(x) \equiv 0 \quad (1)$$

for values of x lying in the $n - 1$ open intervals

$$(x_1, x_2), (x_2, x_3), \dots, (x_{n-1}, x_n)$$

between the n points.

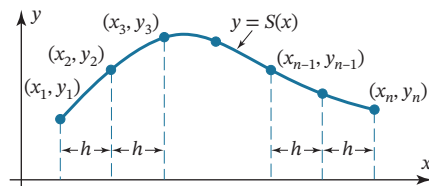


FIGURE 10.3.5

We also need the result from linear beam theory that states that for a beam acted upon only by external forces, the displacement must have two continuous derivatives. In the case of the interpolating curve $y = S(x)$ constructed by the drafting spline, this means that $S(x)$, $S'(x)$, and $S''(x)$ must be continuous for $x_1 \leq x \leq x_n$.

The condition that $S''(x)$ be continuous is what causes a drafting spline to produce a pleasing curve, as it results in continuous *curvature*. The eye can perceive sudden changes in curvature—that is, discontinuities in $S''(x)$ —but sudden changes in higher derivatives

are not discernible. Thus, the condition that $S''(x)$ be continuous is the minimal prerequisite for the interpolating curve to be perceptible as a single smooth curve, rather than as a series of separate curves pieced together.

To determine the mathematical form of the function $S(x)$, we observe that because $S^{(iv)}(x) \equiv 0$ in the intervals between the n specified points, it follows by integrating this equation four times that $S(x)$ must be a *cubic polynomial* in x in each such interval. In general, however, $S(x)$ will be a different cubic polynomial in each interval, so $S(x)$ must have the form

$$S(x) = \begin{cases} S_1(x), & x_1 \leq x \leq x_2 \\ S_2(x), & x_2 \leq x \leq x_3 \\ \vdots & \\ S_{n-1}(x), & x_{n-1} \leq x \leq x_n \end{cases} \quad (2)$$

where $S_1(x), S_2(x), \dots, S_{n-1}(x)$ are cubic polynomials. For convenience, we will write these in the form

$$\begin{aligned} S_1(x) &= a_1(x-x_1)^3 + b_1(x-x_1)^2 + c_1(x-x_1) + d_1, & x_1 \leq x \leq x_2 \\ S_2(x) &= a_2(x-x_2)^3 + b_2(x-x_2)^2 + c_2(x-x_2) + d_2, & x_2 \leq x \leq x_3 \\ &\vdots & \\ S_{n-1}(x) &= a_{n-1}(x-x_{n-1})^3 + b_{n-1}(x-x_{n-1})^2 \\ &\quad + c_{n-1}(x-x_{n-1}) + d_{n-1}, & x_{n-1} \leq x \leq x_n \end{aligned} \quad (3)$$

The a_i 's, b_i 's, c_i 's, and d_i 's constitute a total of $4n - 4$ coefficients that we must determine to specify $S(x)$ completely. If we choose these coefficients so that $S(x)$ interpolates the n specified points in the plane and $S(x)$, $S'(x)$, and $S''(x)$ are continuous, then the resulting interpolating curve is called a **cubic spline**.

Derivation of the Formula of a Cubic Spline

From Equations (2) and (3), we have

$$\begin{aligned} S(x) &= S_1(x) = a_1(x-x_1)^3 + b_1(x-x_1)^2 + c_1(x-x_1) + d_1, & x_1 \leq x \leq x_2 \\ S(x) &= S_2(x) = a_2(x-x_2)^3 + b_2(x-x_2)^2 + c_2(x-x_2) + d_2, & x_2 \leq x \leq x_3 \\ &\vdots & \\ S(x) &= S_{n-1}(x) = a_{n-1}(x-x_{n-1})^3 + b_{n-1}(x-x_{n-1})^2 \\ &\quad + c_{n-1}(x-x_{n-1}) + d_{n-1}, & x_{n-1} \leq x \leq x_n \end{aligned} \quad (4)$$

so

$$\begin{aligned} S'(x) &= S'_1(x) = 3a_1(x-x_1)^2 + 2b_1(x-x_1) + c_1, & x_1 \leq x \leq x_2 \\ S'(x) &= S'_2(x) = 3a_2(x-x_2)^2 + 2b_2(x-x_2) + c_2, & x_2 \leq x \leq x_3 \\ &\vdots & \\ S'(x) &= S'_{n-1}(x) = 3a_{n-1}(x-x_{n-1})^2 + 2b_{n-1}(x-x_{n-1}) + c_{n-1}, & x_{n-1} \leq x \leq x_n \end{aligned} \quad (5)$$

and

$$\begin{aligned} S''(x) &= S''_1(x) = 6a_1(x-x_1) + 2b_1, & x_1 \leq x \leq x_2 \\ S''(x) &= S''_2(x) = 6a_2(x-x_2) + 2b_2, & x_2 \leq x \leq x_3 \\ &\vdots & \\ S''(x) &= S''_{n-1}(x) = 6a_{n-1}(x-x_{n-1}) + 2b_{n-1}, & x_{n-1} \leq x \leq x_n \end{aligned} \quad (6)$$

We will now use these equations and the four properties of cubic splines stated below to express the unknown coefficients a_i , b_i , c_i , d_i , $i = 1, 2, \dots, n-1$, in terms of the known coordinates $y_1, y_2, \dots, y_n$.

1. $S(x)$ interpolates the points (x_i, y_i) , $i = 1, 2, \dots, n$.

Because $S(x)$ interpolates the points (x_i, y_i) , $i = 1, 2, \dots, n$, we have

$$S(x_1) = y_1, \quad S(x_2) = y_2, \dots, \quad S(x_n) = y_n \quad (7)$$

From the first $n - 1$ of these equations and (4), we obtain

$$\begin{aligned} d_1 &= y_1 \\ d_2 &= y_2 \\ &\vdots \\ d_{n-1} &= y_{n-1} \end{aligned} \quad (8)$$

From the last equation in (7), the last equation in (4), and the fact that $x_n - x_{n-1} = h$, we obtain

$$a_{n-1}h^3 + b_{n-1}h^2 + c_{n-1}h + d_{n-1} = y_n \quad (9)$$

2. $S(x)$ is continuous on $[x_1, x_n]$.

Because $S(x)$ is continuous for $x_1 \leq x \leq x_n$, it follows that at each point x_i in the set $x_2, x_3, \dots, x_{n-1}$ we must have

$$S_{i-1}(x_i) = S_i(x_i), \quad i = 2, 3, \dots, n-1 \quad (10)$$

Otherwise, the graphs of $S_{i-1}(x)$ and $S_i(x)$ would not join together to form a continuous curve at x_i . When we apply the interpolating property $S_i(x_i) = y_i$, it follows from (10) that $S_{i-1}(x_i) = y_i$, $i = 2, 3, \dots, n-1$, or from (4) that

$$\begin{aligned} a_1h^3 + b_1h^2 + c_1h + d_1 &= y_2 \\ a_2h^3 + b_2h^2 + c_2h + d_2 &= y_3 \\ &\vdots \\ a_{n-2}h^3 + b_{n-2}h^2 + c_{n-2}h + d_{n-2} &= y_{n-1} \end{aligned} \quad (11)$$

3. $S'(x)$ is continuous on $[x_1, x_n]$.

Because $S'(x)$ is continuous for $x_1 \leq x \leq x_n$, it follows that

$$S'_{i-1}(x_i) = S'_i(x_i), \quad i = 2, 3, \dots, n-1$$

or, from (5),

$$\begin{aligned} 3a_1h^2 + 2b_1h + c_1 &= c_2 \\ 3a_2h^2 + 2b_2h + c_2 &= c_3 \\ &\vdots \\ 3a_{n-2}h^2 + 2b_{n-2}h + c_{n-2} &= c_{n-1} \end{aligned} \quad (12)$$

4. $S''(x)$ is continuous on $[x_1, x_2]$.

Because $S''(x)$ is continuous for $x_1 \leq x \leq x_n$, it follows that

$$S''_{i-1}(x_i) = S''_i(x_i), \quad i = 2, 3, \dots, n-1$$

or, from (6),

$$\begin{aligned} 6a_1h + 2b_1 &= 2b_2 \\ 6a_2h + 2b_2 &= 2b_3 \\ &\vdots \\ 6a_{n-2}h + 2b_{n-2} &= 2b_{n-1} \end{aligned} \quad (13)$$

Equations (8), (9), (11), (12), and (13) constitute a system of $4n - 6$ linear equations in the $4n - 4$ unknown coefficients $a_i, b_i, c_i, d_i, i = 1, 2, \dots, n-1$. Consequently, we need two more equations to determine these coefficients uniquely. Before obtaining these additional equations, however, we can simplify our existing system by expressing the unknowns a_i, b_i, c_i , and d_i in terms of new unknown quantities

$$M_1 = S''(x_1), \quad M_2 = S''(x_2), \dots, \quad M_n = S''(x_n)$$

and the known quantities

$$y_1, y_2, \dots, y_n$$

For example, from (6) it follows that

$$\begin{aligned} M_1 &= 2b_1 \\ M_2 &= 2b_2 \\ &\vdots \\ M_{n-1} &= 2b_{n-1} \end{aligned}$$

so

$$b_1 = \frac{1}{2}M_1, \quad b_2 = \frac{1}{2}M_2, \dots, \quad b_{n-1} = \frac{1}{2}M_{n-1}$$

Moreover, we already know from (8) that

$$d_1 = y_1, \quad d_2 = y_2, \dots, \quad d_{n-1} = y_{n-1}$$

We leave it as an exercise for you to derive the expressions for the a_i 's and c_i 's in terms of the M_i 's and y_i 's. The final result is as follows:

Theorem 10.3.1

Cubic Spline Interpolation

Given n points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ with $x_{i+1} - x_i = h$, $i = 1, 2, \dots, n-1$, the cubic spline

$$S(x) = \begin{cases} a_1(x-x_1)^3 + b_1(x-x_1)^2 + c_1(x-x_1) + d_1, & x_1 \leq x \leq x_2 \\ a_2(x-x_2)^3 + b_2(x-x_2)^2 + c_2(x-x_2) + d_2, & x_2 \leq x \leq x_3 \\ \vdots & \\ a_{n-1}(x-x_{n-1})^3 + b_{n-1}(x-x_{n-1})^2 \\ \quad + c_{n-1}(x-x_{n-1}) + d_{n-1}, & x_{n-1} \leq x \leq x_n \end{cases}$$

that interpolates these points has coefficients given by

$$\begin{aligned} a_i &= (M_{i+1} - M_i)/6h \\ b_i &= M_i/2 \\ c_i &= (y_{i+1} - y_i)/h - [(M_{i+1} + 2M_i)h/6] \\ d_i &= y_i \end{aligned} \tag{14}$$

for $i = 1, 2, \dots, n-1$, where $M_i = S''(x_i)$, $i = 1, 2, \dots, n$.

From this result, we see that the quantities $M_1, M_2, \dots, M_n$ uniquely determine the cubic spline. To find these quantities, we substitute the expressions for a_i , b_i , and c_i given in (14) into (12). After some algebraic simplification, we obtain

$$\begin{aligned} M_1 + 4M_2 + M_3 &= 6(y_1 - 2y_2 + y_3)/h^2 \\ M_2 + 4M_3 + M_4 &= 6(y_2 - 2y_3 + y_4)/h^2 \\ &\vdots \\ M_{n-2} + 4M_{n-1} + M_n &= 6(y_{n-2} - 2y_{n-1} + y_n)/h^2 \end{aligned} \tag{15}$$

or, in matrix form,

$$\begin{bmatrix} 1 & 4 & 1 & 0 & \cdots & 0 & 0 & 0 & 0 \\ 0 & 1 & 4 & 1 & \cdots & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 4 & \cdots & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 4 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 1 & 4 & 1 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & 4 & 1 \end{bmatrix} \begin{bmatrix} M_1 \\ M_2 \\ M_3 \\ M_4 \\ \vdots \\ M_{n-3} \\ M_{n-2} \\ M_{n-1} \\ M_n \end{bmatrix} = \frac{6}{h^2} \begin{bmatrix} y_1 - 2y_2 + y_3 \\ y_2 - 2y_3 + y_4 \\ y_3 - 2y_4 + y_5 \\ \vdots \\ y_{n-4} - 2y_{n-3} + y_{n-2} \\ y_{n-3} - 2y_{n-2} + y_{n-1} \\ y_{n-2} - 2y_{n-1} + y_n \end{bmatrix}$$

This is a linear system of $n - 2$ equations for the n unknowns $M_1, M_2, \dots, M_n$. Thus, we still need two additional equations to determine $M_1, M_2, \dots, M_n$ uniquely. The reason for this is that there are infinitely many cubic splines that interpolate the given points, so we simply do not have enough conditions to determine a unique cubic spline passing through the points. We discuss below three possible ways of specifying the two additional conditions required to obtain a unique cubic spline through the points. (The exercises present two more.) They are summarized in [Table 1](#).

TABLE 1

Natural Spline	The second derivative of the spline is zero at the endpoints.	$M_1 = 0$ $M_n = 0$	$\begin{bmatrix} 4 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 4 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 4 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} M_2 \\ M_3 \\ \vdots \\ M_{n-2} \\ M_{n-1} \end{bmatrix} = \frac{6}{h^2} \begin{bmatrix} y_1 - 2y_2 + y_3 \\ y_2 - 2y_3 + y_4 \\ \vdots \\ y_{n-2} - 2y_{n-1} + y_n \end{bmatrix}$
Parabolic Runout Spline	The spline reduces to a parabolic curve on the first and last intervals.	$M_1 = M_2$ $M_n = M_{n-1}$	$\begin{bmatrix} 5 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 4 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 4 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 5 \end{bmatrix} \begin{bmatrix} M_2 \\ M_3 \\ \vdots \\ M_{n-2} \\ M_{n-1} \end{bmatrix} = \frac{6}{h^2} \begin{bmatrix} y_1 - 2y_2 + y_3 \\ y_2 - 2y_3 + y_4 \\ \vdots \\ y_{n-2} - 2y_{n-1} + y_n \end{bmatrix}$
Cubic Runout Spline	The spline is a single cubic curve on the first two and last two intervals.	$M_1 = 2M_2 - M_3$ $M_n = 2M_{n-1} - M_{n-2}$	$\begin{bmatrix} 6 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 4 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 4 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 6 \end{bmatrix} \begin{bmatrix} M_2 \\ M_3 \\ \vdots \\ M_{n-2} \\ M_{n-1} \end{bmatrix} = \frac{6}{h^2} \begin{bmatrix} y_1 - 2y_2 + y_3 \\ y_2 - 2y_3 + y_4 \\ \vdots \\ y_{n-2} - 2y_{n-1} + y_n \end{bmatrix}$

The Natural Spline

The two simplest mathematical conditions we can impose are

$$M_1 = M_n = 0$$

These conditions together with (15) result in an $n \times n$ linear system for $M_1, M_2, \dots, M_n$, which can be written in matrix form as

$$\begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 4 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 4 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & 4 & 1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} M_1 \\ M_2 \\ M_3 \\ \vdots \\ M_{n-1} \\ M_n \end{bmatrix} = \frac{6}{h^2} \begin{bmatrix} 0 \\ y_1 - 2y_2 + y_3 \\ y_2 - 2y_3 + y_4 \\ \vdots \\ y_{n-2} - 2y_{n-1} + y_n \\ 0 \end{bmatrix}$$

For numerical calculations it is more convenient to eliminate M_1 and M_n from this system and write

$$\begin{bmatrix} 4 & 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 4 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 4 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & 4 & 1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} M_2 \\ M_3 \\ M_4 \\ \vdots \\ M_{n-2} \\ M_{n-1} \end{bmatrix} = \frac{6}{h^2} \begin{bmatrix} y_1 - 2y_2 + y_3 \\ y_2 - 2y_3 + y_4 \\ y_3 - 2y_4 + y_5 \\ \vdots \\ y_{n-3} - 2y_{n-2} + y_{n-1} \\ y_{n-2} - 2y_{n-1} + y_n \end{bmatrix} \quad (16)$$

together with

$$M_1 = 0 \quad (17)$$

$$M_n = 0 \quad (18)$$

Thus, the $(n-2) \times (n-2)$ linear system can be solved for the $n-2$ coefficients $M_2, M_3, \dots, M_{n-1}$, and M_1 and M_n are determined by (17) and (18).

Physically, the natural spline results when the ends of a drafting spline extend freely beyond the interpolating points without constraint. The end portions of the spline outside the interpolating points will fall on straight line paths, causing $S''(x)$ to vanish at the endpoints x_1 and x_n and resulting in the mathematical conditions $M_1 = M_n = 0$.

The natural spline tends to flatten the interpolating curve at the endpoints, which may be undesirable. Of course, if it is required that $S''(x)$ vanish at the endpoints, then the natural spline must be used.

The Parabolic Runout Spline

The two additional constraints imposed for this type of spline are

$$M_1 = M_2 \quad (19)$$

$$M_n = M_{n-1} \quad (20)$$

If we use the preceding two equations to eliminate M_1 and M_n from (15), we obtain the $(n-2) \times (n-2)$ linear system

$$\begin{bmatrix} 5 & 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 4 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 4 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & 4 & 1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & 5 \end{bmatrix} \begin{bmatrix} M_2 \\ M_3 \\ M_4 \\ \vdots \\ M_{n-2} \\ M_{n-1} \end{bmatrix} = \frac{6}{h^2} \begin{bmatrix} y_1 - 2y_2 + y_3 \\ y_2 - 2y_3 + y_4 \\ y_3 - 2y_4 + y_5 \\ \vdots \\ y_{n-3} - 2y_{n-2} + y_{n-1} \\ y_{n-2} - 2y_{n-1} + y_n \end{bmatrix} \quad (21)$$

for $M_2, M_3, \dots, M_{n-1}$. Once these $n-2$ values have been determined, M_1 and M_n are determined from (19) and (20).

From (14) we see that $M_1 = M_2$ implies that $a_1 = 0$, and $M_n = M_{n-1}$ implies that $a_{n-1} = 0$. Thus, from (3) there are no cubic terms in the formula for the spline over the end intervals $[x_1, x_2]$ and $[x_{n-1}, x_n]$. Hence, as the name suggests, the parabolic runout spline reduces to a parabolic curve over these end intervals.

The Cubic Runout Spline

For this type of spline, we impose the two additional conditions

$$M_1 = 2M_2 - M_3 \quad (22)$$

$$M_n = 2M_{n-1} - M_{n-2} \quad (23)$$

Using these two equations to eliminate M_1 and M_n from (15) results in the following $(n-2) \times (n-2)$ linear system for $M_2, M_3, \dots, M_{n-1}$:

$$\begin{bmatrix} 6 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 4 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 4 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & 4 & 1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} M_2 \\ M_3 \\ M_4 \\ \vdots \\ M_{n-2} \\ M_{n-1} \end{bmatrix} = \frac{6}{h^2} \begin{bmatrix} y_1 - 2y_2 + y_3 \\ y_2 - 2y_3 + y_4 \\ y_3 - 2y_4 + y_5 \\ \vdots \\ y_{n-3} - 2y_{n-2} + y_{n-1} \\ y_{n-2} - 2y_{n-1} + y_n \end{bmatrix} \quad (24)$$

After we solve this linear system for $M_2, M_3, \dots, M_{n-1}$, we can use (22) and (23) to determine M_1 and M_n .

If we rewrite (22) as

$$M_2 - M_1 = M_3 - M_2$$

it follows from (14) that $a_1 = a_2$. Because $S'''(x) = 6a_1$ on $[x_1, x_2]$ and $S'''(x) = 6a_2$ on $[x_2, x_3]$, we see that $S'''(x)$ is constant over the entire interval $[x_1, x_3]$. Consequently, $S(x)$ consists of a *single* cubic curve over the interval $[x_1, x_3]$ rather than two different cubic

curves pieced together at x_2 . [To see this, integrate $S'''(x)$ three times.] A similar analysis shows that $S(x)$ consists of a single cubic curve over the last two intervals.

Whereas the natural spline tends to produce an interpolating curve that is flat at the endpoints, the cubic runout spline has the opposite tendency: it produces a curve with pronounced curvature at the endpoints. If neither behavior is desired, the parabolic runout spline is a reasonable compromise.

EXAMPLE 1 | Using a Parabolic Runout Spline

The density of water is well known to reach a maximum at a temperature slightly above freezing. **Table 2**, from the *Handbook of Chemistry and Physics* (CRC Press, 2009), gives the density of water in grams per cubic centimeter for five equally spaced temperatures from -10°C to 30°C . We will interpolate these five temperature–density measurements with a parabolic runout spline and attempt to find the maximum density of water in this range by finding the maximum value on this cubic spline. In the exercises we ask you to perform similar calculations using a natural spline and a cubic runout spline to interpolate the data points.

Set

$$\begin{aligned}x_1 &= -10, & y_1 &= .99815 \\x_2 &= 0, & y_2 &= .99987 \\x_3 &= 10, & y_3 &= .99973 \\x_4 &= 20, & y_4 &= .99823 \\x_5 &= 30, & y_5 &= .99567\end{aligned}$$

Then

$$\begin{aligned}6[y_1 - 2y_2 + y_3]/h^2 &= -.0001116 \\6[y_2 - 2y_3 + y_4]/h^2 &= -.0000816 \\6[y_3 - 2y_4 + y_5]/h^2 &= -.0000636\end{aligned}$$

and the linear system (21) for the parabolic runout spline becomes

$$\begin{bmatrix} 5 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 5 \end{bmatrix} \begin{bmatrix} M_2 \\ M_3 \\ M_4 \end{bmatrix} = \begin{bmatrix} -.0001116 \\ -.0000816 \\ -.0000636 \end{bmatrix}$$

Solving this system yields

$$\begin{aligned}M_2 &= -.00001973 \\M_3 &= -.00001293 \\M_4 &= -.00001013\end{aligned}$$

From (19) and (20), we have

$$\begin{aligned}M_1 &= M_2 = -.00001973 \\M_5 &= M_4 = -.00001013\end{aligned}$$

Solving for the a_i 's, b_i 's, c_i 's, and d_i 's in (14), we obtain the following expression for the interpolating parabolic runout spline:

$$S(x) = \begin{cases} -.00000987(x+10)^2 + .0002707(x+10) + .99815, & -10 \leq x \leq 0 \\ .000000113(x-0)^3 -.00000987(x-0)^2 + .0000733(x-0) + .99987, & 0 \leq x \leq 10 \\ .000000047(x-10)^3 -.00000647(x-10)^2 -.0000900(x-10) + .99973, & 10 \leq x \leq 20 \\ -.00000507(x-20)^2 -.0002053(x-20) + .99823, & 20 \leq x \leq 30 \end{cases}$$

This spline is plotted in **Figure 10.3.6**. From that figure we see that the maximum is attained in the interval $[0, 10]$. To find this maximum, we set $S'(x)$ equal to zero in the interval $[0, 10]$:

$$S'(x) = .000000339x^2 - .0000197x + .0000733 = 0$$

To three significant digits the root of this quadratic in the interval $[0, 10]$ is $x = 3.99$, and for this value of x , $S(3.99) = 1.00001$. Thus, according to our interpolated estimate, the maximum density of water is 1.00001 g/cm^3 attained at 3.99°C . This agrees well with the experimental maximum density of 1.00000 g/cm^3 attained at 3.98°C . (In the original metric system, the gram was *defined* as the mass of one cubic centimeter of water at its maximum density.)

TABLE 2

Temperature ($^{\circ}\text{C}$)	Density (g/cm^3)
-10	.99815
0	.99987
10	.99973
20	.99823
30	.99567

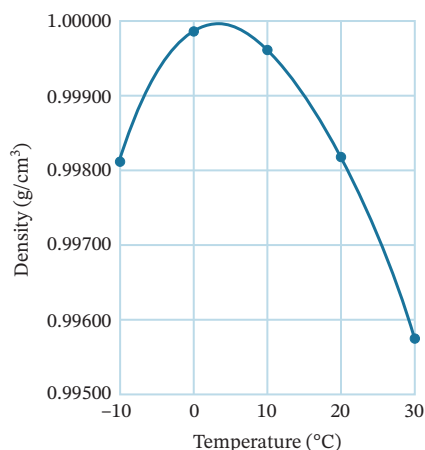


FIGURE 10.3.6

Closing Remarks

In addition to producing excellent interpolating curves, cubic splines and their generalizations are useful for numerical integration and differentiation, for the numerical solution of differential and integral equations, and in optimization theory.

Exercise Set 10.3

- Derive the expressions for a_i and c_i in Equations (14) of Theorem 10.3.1.
- The six points

$$(0, .00000), \quad (.2, .19867), \quad (.4, .38942), \\ (.6, .56464), \quad (.8, .71736), \quad (1.0, .84147)$$

lie on the graph of $y = \sin x$, where x is in radians.

- Find the portion of the parabolic runout spline that interpolates these six points for $.4 \leq x \leq .6$. Maintain an accuracy of five decimal places in your calculations.
 - Calculate $S(.5)$ for the spline you found in part (a). What is the percentage error of $S(.5)$ with respect to the “exact” value of $\sin(.5) = .47943$?
- The following five points

$$(0, 1), \quad (1, 7), \quad (2, 27), \quad (3, 79), \quad (4, 181)$$

lie on a single cubic curve.

- Which of the three types of cubic splines (natural, parabolic runout, or cubic runout) would agree exactly with the single cubic curve on which the five points lie?
 - Determine the cubic spline you chose in part (a), and verify that it is a single cubic curve that interpolates the five points.
- Repeat the calculations in Example 1 using a natural spline to interpolate the five data points.
 - Repeat the calculations in Example 1 using a cubic runout spline to interpolate the five data points.
 - Consider the five points $(0, 0)$, $(.5, 1)$, $(1, 0)$, $(1.5, -1)$, and $(2, 0)$ on the graph of $y = \sin(\pi x)$.
 - Use a natural spline to interpolate the data points $(0, 0)$, $(.5, 1)$, and $(1, 0)$.

- b. Use a natural spline to interpolate the data points $(.5, 1)$, $(1, 0)$, and $(1.5, -1)$.
- c. Explain the unusual nature of your result in part (b).

7. **(The Periodic Spline)** If it is known or if it is desired that the n points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ to be interpolated lie on a single cycle of a periodic curve with period $x_n - x_1$, then an interpolating cubic spline $S(x)$ must satisfy

$$\begin{aligned} S(x_1) &= S(x_n) \\ S'(x_1) &= S'(x_n) \\ S''(x_1) &= S''(x_n) \end{aligned}$$

- a. Show that these three periodicity conditions require that

$$\begin{aligned} y_1 &= y_n \\ M_1 &= M_n \\ 4M_1 + M_2 + M_{n-1} &= 6(y_{n-1} - 2y_1 + y_2)/h^2 \end{aligned}$$

- b. Using the three equations in part (a) and Equations (15), construct an $(n-1) \times (n-1)$ linear system for $M_1, M_2, \dots, M_{n-1}$ in matrix form.

8. **(The Clamped Spline)** Suppose that, in addition to the n points to be interpolated, we are given specific values y'_1 and y'_n for the slopes $S'(x_1)$ and $S'(x_n)$ of the interpolating cubic spline at the endpoints x_1 and x_n .

- a. Show that

$$\begin{aligned} 2M_1 + M_2 &= 6(y_2 - y_1 - hy'_1)/h^2 \\ 2M_n + M_{n-1} &= 6(y_{n-1} - y_n + hy'_n)/h^2 \end{aligned}$$

- b. Using the equations in part (a) and Equations (15), construct an $n \times n$ linear system for $M_1, M_2, \dots, M_n$ in matrix form.

Remark The clamped spline described in this exercise is the most accurate type of spline for interpolation work if the slopes at the endpoints are known or can be estimated.

Working with Technology

The following exercises are designed to be solved using a technology utility. Typically, this will be MATLAB, Mathematica, Maple, Derive, or Mathcad, but it may also be some other type of linear algebra software or a scientific calculator with some linear algebra capabilities. For each exercise you will need to read the relevant documentation for the particular utility you are using. The goal of these exercises is to provide you with a basic proficiency with your technology utility. Once you have mastered the techniques in these exercises, you will be able to use your technology utility to solve many of the problems in the regular exercise sets.

- T1. In the solution of the natural cubic spline problem, it is necessary to solve a system of equations having coefficient matrix

$$A_n = \begin{bmatrix} 4 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 4 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 4 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 4 \end{bmatrix}$$

If we can present a formula for the inverse of this matrix, then the solution for the natural cubic spline problem can be easily

obtained. In this exercise and the next, we use a computer to discover this formula. Toward this end, we first determine an expression for the determinant of A_n , denoted by the symbol D_n . Given that

$$A_1 = [4] \quad \text{and} \quad A_2 = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$$

we see that

$$D_1 = \det(A_1) = \det[4] = 4$$

and

$$D_2 = \det(A_2) = \det \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix} = 15$$

- a. Use the cofactor expansion of determinants to show that

$$D_n = 4D_{n-1} - D_{n-2}$$

for $n = 3, 4, 5, \dots$. This says, for example, that

$$\begin{aligned} D_3 &= 4D_2 - D_1 = 4(15) - 4 = 56 \\ D_4 &= 4D_3 - D_2 = 4(56) - 15 = 209 \end{aligned}$$

and so on. Using a computer, check this result for $5 \leq n \leq 10$.

- b. By writing

$$D_n = 4D_{n-1} - D_{n-2}$$

and the identity, $D_{n-1} = D_{n-1}$, in matrix form,

$$\begin{bmatrix} D_n \\ D_{n-1} \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} D_{n-1} \\ D_{n-2} \end{bmatrix}$$

show that

$$\begin{bmatrix} D_n \\ D_{n-1} \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ 1 & 0 \end{bmatrix}^{n-2} \begin{bmatrix} D_2 \\ D_1 \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ 1 & 0 \end{bmatrix}^{n-2} \begin{bmatrix} 15 \\ 4 \end{bmatrix}$$

- c. Use the methods in Section 5.2 and a computer to show that

$$\begin{bmatrix} 4 & -1 \\ 1 & 0 \end{bmatrix}^{n-2} = \frac{\begin{bmatrix} (2 + \sqrt{3})^{n-1} - (2 - \sqrt{3})^{n-1} & (2 - \sqrt{3})^{n-2} - (2 + \sqrt{3})^{n-2} \\ (2 + \sqrt{3})^{n-2} - (2 - \sqrt{3})^{n-2} & (2 - \sqrt{3})^{n-3} - (2 + \sqrt{3})^{n-3} \end{bmatrix}}{2\sqrt{3}}$$

and hence

$$D_n = \frac{(2 + \sqrt{3})^{n+1} - (2 - \sqrt{3})^{n+1}}{2\sqrt{3}}$$

for $n = 1, 2, 3, \dots$

- d. Using a computer, check this result for $1 \leq n \leq 10$.

- T2. In this exercise, we determine a formula for calculating A_n^{-1} from D_k for $k = 0, 1, 2, 3, \dots, n$, assuming that D_0 is defined to be 1.

- a. Use a computer to compute A_k^{-1} for $k = 1, 2, 3, 4$, and 5.
- b. From your results in part (a), discover the conjecture that

$$A_n^{-1} = [\alpha_{ij}]$$

where $\alpha_{ij} = \alpha_{ji}$ and

$$\alpha_{ij} = (-1)^{i+j} \left(\frac{D_{n-j} D_{i-1}}{D_n} \right)$$

for $i \leq j$.

- c. Use the result in part (b) to compute A_7^{-1} and compare it to the result obtained using the computer.

10.4

Markov Chains

In this section we describe a general model of a system that changes from state to state. We then apply the model to several concrete problems.

PREREQUISITES:

Linear Systems
 Matrices
 Intuitive Understanding of Limits

A Markov Process

Suppose a physical or mathematical system undergoes a process of change such that at any moment it can occupy one of a finite number of states. For example, the weather in a certain city could be in one of three possible states: sunny, cloudy, or rainy. Or an individual could be in one of four possible emotional states: happy, sad, angry, or apprehensive. Suppose that such a system changes with time from one state to another and at scheduled times the state of the system is observed. If the state of the system at any observation cannot be predicted with certainty, but the probability that a given state occurs can be predicted by just knowing the state of the system at the preceding observation, then the process of change is called a **Markov chain** or **Markov process**.

Definition 1

If a Markov chain has k possible states, which we label as $1, 2, \dots, k$, then the probability that the system is in state i at any observation after it was in state j at the preceding observation is denoted by p_{ij} and is called the **transition probability** from state j to state i . The matrix $P = [p_{ij}]$ is called the **transition matrix of the Markov chain**.

For example, in a three-state Markov chain, the transition matrix has the form

Preceding State

1

2

3

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix}$$

1

2

3

New State

In this matrix, p_{32} is the probability that the system will change from state 2 to state 3, p_{11} is the probability that the system will still be in state 1 if it was previously in state 1, and so forth.

EXAMPLE 1 | Transition Matrix of the Markov Chain

A car rental agency has three rental locations, denoted by 1, 2, and 3. A customer may rent a car from any of the three locations and return the car to any of the three locations. The manager finds that customers return the cars to the various locations according to the following probabilities:

Rented from Location

1

2

3

$$\begin{bmatrix} .8 & .3 & .2 \\ .1 & .2 & .6 \\ .1 & .5 & .2 \end{bmatrix}$$

1

2

3

Returned to Location

This matrix is the transition matrix of the system considered as a Markov chain. From this matrix, the probability is .6 that a car rented from location 3 will be returned to location 2, the probability is .8 that a car rented from location 1 will be returned to location 1, and so forth.

EXAMPLE 2 | Transition Matrix of the Markov Chain

By reviewing its donation records, the alumni office of a college finds that 80% of its alumni who contribute to the annual fund one year will also contribute the next year, and 30% of those who do not contribute one year will contribute the next. This can be viewed as a Markov chain with two states: state 1 corresponds to an alumnus giving a donation in any one year, and state 2 corresponds to the alumnus not giving a donation in that year. The transition matrix is

$$P = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$$

In the preceding examples, the transition matrices of the Markov chains have the property that the entries in any column sum to 1. This is not accidental. If $P = [p_{ij}]$ is the transition matrix of any Markov chain with k states, then for each j we must have

$$p_{1j} + p_{2j} + \cdots + p_{kj} = 1 \quad (1)$$

because if the system is in state j at one observation, it is certain to be in one of the k possible states at the next observation.

A matrix with property (1) is called a **stochastic matrix**, a **probability matrix**, or a **Markov matrix**. From the preceding discussion, it follows that the transition matrix for a Markov chain must be a stochastic matrix.

In a Markov chain, the state of the system at any observation time cannot generally be determined with certainty. The best one can usually do is specify probabilities for each of the possible states. For example, in a Markov chain with three states, we might describe the possible state of the system at some observation time by a column vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

in which x_1 is the probability that the system is in state 1, x_2 the probability that it is in state 2, and x_3 the probability that it is in state 3. In general we make the following definition.

Definition 2

The **state vector** for an observation of a Markov chain with k states is a column vector $\mathbf{x}$ whose i th component x_i is the probability that the system is in the i th state at that time.

Observe that the entries in any state vector for a Markov chain are nonnegative and have a sum of 1. (Why?) A column vector that has this property is called a **probability vector**.

Let us suppose now that we know the state vector $\mathbf{x}^{(0)}$ for a Markov chain at some initial observation. The following theorem will enable us to determine the state vectors

$$\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(n)}, \dots$$

at the subsequent observation times.

Theorem 10.4.1

If P is the transition matrix of a Markov chain and $\mathbf{x}^{(n)}$ is the state vector at the n th observation, then $\mathbf{x}^{(n+1)} = P\mathbf{x}^{(n)}$.

The proof of this theorem involves ideas from probability theory and will not be given here. From this theorem, it follows that

$$\begin{aligned}\mathbf{x}^{(1)} &= P\mathbf{x}^{(0)} \\ \mathbf{x}^{(2)} &= P\mathbf{x}^{(1)} = P^2\mathbf{x}^{(0)} \\ \mathbf{x}^{(3)} &= P\mathbf{x}^{(2)} = P^3\mathbf{x}^{(0)} \\ &\vdots \\ \mathbf{x}^{(n)} &= P\mathbf{x}^{(n-1)} = P^n\mathbf{x}^{(0)}\end{aligned}$$

In this way, the initial state vector $\mathbf{x}^{(0)}$ and the transition matrix P determine $\mathbf{x}^{(n)}$ for $n = 1, 2, \dots$.

EXAMPLE 3 | Example 2 Revisited

The transition matrix in Example 2 was

$$P = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$$

We now construct the probable future donation record of a new graduate who did not give a donation in the initial year after graduation. For such a graduate the system is initially in state 2 with certainty, so the initial state vector is

$$\mathbf{x}^{(0)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

From Theorem 10.4.1 we then have

$$\begin{aligned}\mathbf{x}^{(1)} &= P\mathbf{x}^{(0)} = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} .3 \\ .7 \end{bmatrix} \\ \mathbf{x}^{(2)} &= P\mathbf{x}^{(1)} = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \begin{bmatrix} .3 \\ .7 \end{bmatrix} = \begin{bmatrix} .45 \\ .55 \end{bmatrix} \\ \mathbf{x}^{(3)} &= P\mathbf{x}^{(2)} = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \begin{bmatrix} .45 \\ .55 \end{bmatrix} = \begin{bmatrix} .525 \\ .475 \end{bmatrix}\end{aligned}$$

Thus, after three years the alumnus can be expected to make a donation with probability .525. Beyond three years, we find the following state vectors (to three decimal places):

$$\begin{aligned}\mathbf{x}^{(4)} &= \begin{bmatrix} .563 \\ .438 \end{bmatrix}, & \mathbf{x}^{(5)} &= \begin{bmatrix} .581 \\ .419 \end{bmatrix}, & \mathbf{x}^{(6)} &= \begin{bmatrix} .591 \\ .409 \end{bmatrix}, & \mathbf{x}^{(7)} &= \begin{bmatrix} .595 \\ .405 \end{bmatrix} \\ \mathbf{x}^{(8)} &= \begin{bmatrix} .598 \\ .402 \end{bmatrix}, & \mathbf{x}^{(9)} &= \begin{bmatrix} .599 \\ .401 \end{bmatrix}, & \mathbf{x}^{(10)} &= \begin{bmatrix} .599 \\ .401 \end{bmatrix}, & \mathbf{x}^{(11)} &= \begin{bmatrix} .600 \\ .400 \end{bmatrix}\end{aligned}$$

For all n beyond 11, we have

$$\mathbf{x}^{(n)} = \begin{bmatrix} .600 \\ .400 \end{bmatrix}$$

to three decimal places. In other words, the state vectors converge to a fixed vector as the number of observations increases. (We will discuss this further next.)

EXAMPLE 4 | Example 1 Revisited

The transition matrix in Example 1 was

$$\begin{bmatrix} .8 & .3 & .2 \\ .1 & .2 & .6 \\ .1 & .5 & .2 \end{bmatrix}$$

If a car is rented initially from location 2, then the initial state vector is

$$\mathbf{x}^{(0)} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Using this vector and Theorem 10.4.1, one obtains the later state vectors listed in **Table 1**.

For all values of n greater than 11, all state vectors are equal to $\mathbf{x}^{(11)}$ to three decimal places.

Two things should be observed in this example. First, it was not necessary to know how long a customer kept the car. That is, in a Markov process the time period between observations need not be regular. Second, the state vectors approach a fixed vector as n increases, just as in the first example.

TABLE 1

$\mathbf{x}^{(n)}$ \ n	0	1	2	3	4	5	6	7	8	9	10	11
$x_1^{(n)}$	0	.300	.400	.477	.511	.533	.544	.550	.553	.555	.556	.557
$x_2^{(n)}$	1	.200	.370	.252	.261	.240	.238	.233	.232	.231	.230	.230
$x_3^{(n)}$	0	.500	.230	.271	.228	.227	.219	.217	.215	.214	.214	.213

EXAMPLE 5 | Using Theorem 10.4.1

A traffic officer is assigned to control the traffic at the eight intersections indicated in **Figure 10.4.1**. She is instructed to remain at each intersection for an hour and then to either remain at the same intersection or move to a neighboring intersection. To avoid establishing a pattern, she is told to choose her new intersection on a random basis, with each possible choice equally likely. For example, if she is at intersection 5, her next intersection can be 2, 4, 5, or 8, each with probability $\frac{1}{4}$. Every day she starts at the location where she stopped the day before. The transition matrix for this Markov chain is

		Old Intersection									
		1	2	3	4	5	6	7	8		
New Intersection	1	$\frac{1}{3}$	$\frac{1}{3}$	0	$\frac{1}{5}$	0	0	0	0	1	
	2	$\frac{1}{3}$	$\frac{1}{3}$	0	0	$\frac{1}{4}$	0	0	0	2	
	3	0	0	$\frac{1}{3}$	$\frac{1}{5}$	0	$\frac{1}{3}$	0	0	3	
	4	$\frac{1}{3}$	0	$\frac{1}{3}$	$\frac{1}{5}$	$\frac{1}{4}$	0	$\frac{1}{4}$	0	4	
	5	0	$\frac{1}{3}$	0	$\frac{1}{5}$	$\frac{1}{4}$	0	0	$\frac{1}{3}$	5	
	6	0	0	$\frac{1}{3}$	0	0	$\frac{1}{3}$	$\frac{1}{4}$	0	6	
	7	0	0	0	$\frac{1}{5}$	0	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{3}$	7	
	8	0	0	0	0	$\frac{1}{4}$	0	$\frac{1}{4}$	$\frac{1}{3}$	8	

If the traffic officer begins at intersection 5, her probable locations, hour by hour, are given by the state vectors given in **Table 2**. For all values of n greater than 22, all state vectors are equal to $\mathbf{x}^{(22)}$ to three decimal places. Thus, as with the first two examples, the state vectors approach a fixed vector as n increases.

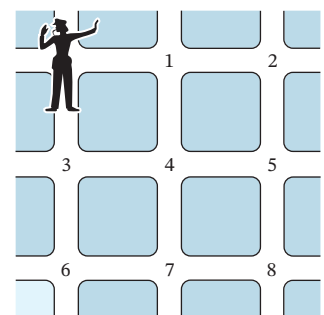
**FIGURE 10.4.1**

TABLE 2

$\mathbf{x}^{(n)} \backslash n$	0	1	2	3	4	5	10	15	20	22
$x_1^{(n)}$	0	.000	.133	.116	.130	.123	.113	.109	.108	.107
$x_2^{(n)}$	0	.250	.146	.163	.140	.138	.115	.109	.108	.107
$x_3^{(n)}$	0	.000	.050	.039	.067	.073	.100	.106	.107	.107
$x_4^{(n)}$	0	.250	.113	.187	.162	.178	.178	.179	.179	.179
$x_5^{(n)}$	1	.250	.279	.190	.190	.168	.149	.144	.143	.143
$x_6^{(n)}$	0	.000	.000	.050	.056	.074	.099	.105	.107	.107
$x_7^{(n)}$	0	.000	.133	.104	.131	.125	.138	.142	.143	.143
$x_8^{(n)}$	0	.250	.146	.152	.124	.121	.108	.107	.107	.107

Limiting Behavior of the State Vectors

In our examples we saw that the state vectors approached some fixed vector as the number of observations increased. We now ask whether the state vectors always approach a fixed vector in a Markov chain. A simple example shows that this is not the case.

EXAMPLE 6 | System Oscillates Between Two State Vectors

Let

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{x}^{(0)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Then, because $P^2 = I$ and $P^3 = P$, we have that

$$\mathbf{x}^{(0)} = \mathbf{x}^{(2)} = \mathbf{x}^{(4)} = \dots = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

and

$$\mathbf{x}^{(1)} = \mathbf{x}^{(3)} = \mathbf{x}^{(5)} = \dots = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

This system oscillates indefinitely between the two state vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, so it does not approach any fixed vector.

However, if we impose a mild condition on the transition matrix, we can show that a fixed limiting state vector is approached. This condition is described by the following definition.

Definition 3

A transition matrix is **regular** if some integer power of it has all positive entries.

Thus, for a regular transition matrix P , there is some positive integer m such that all entries of P^m are positive. This is the case with the transition matrices of Examples 1 and 2 for $m = 1$. In Example 5 it turns out that P^4 has all positive entries. Consequently, in all three examples the transition matrices are regular.

A Markov chain that is governed by a regular transition matrix is called a **regular Markov chain**. We will see that every regular Markov chain has a fixed state vector $\mathbf{q}$ such that $P^n \mathbf{x}^{(0)}$ approaches $\mathbf{q}$ as n increases for any choice of $\mathbf{x}^{(0)}$. This result is of major importance in the theory of Markov chains. It is based on the following theorem.

Theorem 10.4.2

Behavior of P^n as $n \rightarrow \infty$

If P is a regular transition matrix, then as $n \rightarrow \infty$,

$$P^n \rightarrow \begin{bmatrix} q_1 & q_1 & \cdots & q_1 \\ q_2 & q_2 & \cdots & q_2 \\ \vdots & \vdots & & \vdots \\ q_k & q_k & \cdots & q_k \end{bmatrix}$$

where the q_i are positive numbers such that $q_1 + q_2 + \cdots + q_k = 1$.

We will not prove this theorem here. We refer you to a more specialized text, such as J. Kemeny and J. Snell, *Finite Markov Chains* (New York: Springer-Verlag, 1976).

Let us set

$$Q = \begin{bmatrix} q_1 & q_1 & \cdots & q_1 \\ q_2 & q_2 & \cdots & q_2 \\ \vdots & \vdots & & \vdots \\ q_k & q_k & \cdots & q_k \end{bmatrix} \quad \text{and} \quad \mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_k \end{bmatrix}$$

Thus, Q is a transition matrix, all of whose columns are equal to the probability vector $\mathbf{q}$. Q has the property that if $\mathbf{x}$ is any probability vector, then

$$\begin{aligned} Q\mathbf{x} &= \begin{bmatrix} q_1 & q_1 & \cdots & q_1 \\ q_2 & q_2 & \cdots & q_2 \\ \vdots & \vdots & & \vdots \\ q_k & q_k & \cdots & q_k \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{bmatrix} = \begin{bmatrix} q_1 x_1 + q_1 x_2 + \cdots + q_1 x_k \\ q_2 x_1 + q_2 x_2 + \cdots + q_2 x_k \\ \vdots \\ q_k x_1 + q_k x_2 + \cdots + q_k x_k \end{bmatrix} \\ &= (x_1 + x_2 + \cdots + x_k) \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_k \end{bmatrix} = (1)\mathbf{q} = \mathbf{q} \end{aligned}$$

That is, Q transforms any probability vector $\mathbf{x}$ into the fixed probability vector $\mathbf{q}$. This result leads to the following theorem.

Theorem 10.4.3

Behavior of $P^n \mathbf{x}$ as $n \rightarrow \infty$

If P is a regular transition matrix and $\mathbf{x}$ is any probability vector, then as $n \rightarrow \infty$,

$$P^n \mathbf{x} \rightarrow \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_k \end{bmatrix} = \mathbf{q}$$

where $\mathbf{q}$ is a fixed probability vector, independent of n , all of whose entries are positive.

This result holds since Theorem 10.4.2 implies that $P^n \rightarrow Q$ as $n \rightarrow \infty$. This in turn implies that $P^n \mathbf{x} \rightarrow Q\mathbf{x} = \mathbf{q}$ as $n \rightarrow \infty$. Thus, for a regular Markov chain, the system eventually approaches a fixed state vector $\mathbf{q}$. The vector $\mathbf{q}$ is called the **steady-state vector** of the regular Markov chain.

For systems with many states, usually the most efficient technique of computing the steady-state vector $\mathbf{q}$ is simply to calculate $P^n \mathbf{x}$ for some large n . Our examples illustrate this procedure. Each is a regular Markov process, so that convergence to a steady-state vector is ensured. Another way of computing the steady-state vector is to make use of the following theorem.

Theorem 10.4.4

Steady-State Vector

The steady-state vector $\mathbf{q}$ of a regular transition matrix P is the unique probability vector that satisfies the equation $P\mathbf{q} = \mathbf{q}$.

To see this, consider the matrix identity $PP^n = P^{n+1}$. By Theorem 10.4.2, both P^n and P^{n+1} approach Q as $n \rightarrow \infty$. Thus, we have $PQ = Q$. Any one column of this matrix equation gives $P\mathbf{q} = \mathbf{q}$. To show that $\mathbf{q}$ is the only probability vector that satisfies this equation, suppose $\mathbf{r}$ is another probability vector such that $P\mathbf{r} = \mathbf{r}$. Then also $P^n \mathbf{r} = \mathbf{r}$ for $n = 1, 2, \dots$. When we let $n \rightarrow \infty$, Theorem 10.4.3 leads to $\mathbf{q} = \mathbf{r}$.

Theorem 10.4.4 can also be expressed by the statement that the homogeneous linear system

$$(I - P)\mathbf{q} = \mathbf{0}$$

has a unique solution vector $\mathbf{q}$ with nonnegative entries that satisfy the condition

$$q_1 + q_2 + \cdots + q_k = 1$$

We can apply this technique to the computation of the steady-state vectors for our examples.

EXAMPLE 7 | Example 2 Revisited

In Example 2 the transition matrix was

$$P = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$$

so the linear system $(I - P)\mathbf{q} = \mathbf{0}$ is

$$\begin{bmatrix} .2 & -.3 \\ -.2 & .3 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (2)$$

This leads to the single independent equation

$$.2q_1 - .3q_2 = 0$$

or

$$q_1 = 1.5q_2$$

Thus, when we set $q_2 = s$, any solution of (2) is of the form

$$\mathbf{q} = s \begin{bmatrix} 1.5 \\ 1 \end{bmatrix}$$